



Integrating UAVs and RISs in Future Wireless Networks: A Review and Tutorial on IoTs and Vehicular Communications

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Abstract: The rapid evolution of smart cities relies heavily on advancements in wireless communication systems and extensive IoT networks. This paper offers a comprehensive review of the critical role and future potential of integrating unmanned aerial vehicles (UAVs) and reconfigurable intelligent surfaces (RISs) to enhance Internet of Vehicles (IoV) systems within beyond-fifth-generation (B5G) and sixth-generation (6G) networks. We explore the combination of quasi-optical millimeter-wave (mmWave) signals with UAV-enabled, RIS-assisted networks and their applications in urban environments. This review covers essential areas such as channel modeling and position-aware beamforming in dynamic networks, including UAVs and IoVs. Moreover, we investigate UAV navigation and control, emphasizing the development of obstacle-free trajectory designs in dense urban areas while meeting kinodynamic and motion constraints. The emerging potential of RIS-equipped UAVs (RISeUAVs) is highlighted, along with their role in supporting IoVs and in mobile edge computing. Optimization techniques, including convex programming methods and machine learning, are explored to tackle complex challenges, with an emphasis on studying computational complexity and feasibility for real-time operations. Additionally, this review highlights the integrated localization and communication strategies to enhance UAV and autonomous ground vehicle operations. This tutorial-style overview offers insights into the technical challenges and innovative solutions of the next-generation wireless networks in smart cities, with a focus on vehicular communications. Finally, future research directions are outlined.

Keywords: autonomous navigation; intelligent vehicles; Internet of Things (IoT); Internet of Vehicles (IoVs); mmWave communication; reconfigurable intelligent surface (RIS); trajectory optimization; vehicular communication; wireless communication; unmanned aerial vehicles (UAVs)



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1. Introduction

The modernization of human life, characterized by advancements in smart cities, smart grids, smart transportation, etc., increasingly depends on wireless communication and sensor networks built upon the massive Internet of Things (IoT) [1–3]. The growing demand for high-bandwidth wireless networks capable of supporting high data transmission rates has driven the development of technologies beyond fifth-generation (B5G), sixth-generation (6G) [4], and optical (visible light) wireless communication networks (WCNs) [5]. Future WCNs leverage quasi-optic millimeter wave (mmWave) electromagnetic signals to achieve higher carrier frequencies, enabling high-speed, high-data-rate wireless communication [6].

Although mmWave signals offer higher bandwidth, they face significant propagation challenges due to rapid attenuation, fast fading, shadowing, as well as blockage and blind-spot issues [7]. To mitigate these limitations, the concept of massive multi-input multi-output (MIMO) communication has been introduced [8]. MIMO base stations (BSs) are equipped with arrays of mmWave antennas [9], enabling them to enhance signal strength through beamforming. This technique focuses electromagnetic waves to transmit steered signals in line-of-sight (LoS) directions toward user equipment (UE) [10].

However, the massive MIMO concept introduces complexity and reduces sustainability in future WCNs due to the intensive baseband signal processing and active beamforming requirements, which can lead to energy and spectral efficiency challenges [11]. Reconfigurable intelligent surfaces (RISs), which act as passive yet smart mmWave reflectors, offer a solution to enhance both spectral and energy efficiency in future WCNs [12]. Unlike active MIMO systems, RISs do not require radio frequency (RF) chains for signal processing and consume minimal energy for passive beamforming, making them a viable option for more sustainable and less complex network architectures [13].

The significant mobility of unmanned aerial vehicles (UAVs), commonly referred to as drones, has proven advantageous for enhancing WCNs by improving coverage [14], ubiquitousness [15], capacity [16], flexibility, throughput [17], and reliability. UAVs can be utilized for data collection in sensor networks [18] or as enablers for mobile edge computing (MEC) to support remote IoT applications [19]. Among UAV types, quadrotors, which are vertical take-off and landing (VTOL) vehicles [20], are particularly well-suited for deployment in dense urban environments. Their ability to maintain proximity for efficient data collection, from IoT sensors that transmit low power over short distances, makes them ideal for such tasks [21]. Additionally, quadrotors play an essential role in smart city applications, especially in intelligent transportation systems involving intelligent vehicles (IVs) [22]. Their flexibility allows them to support aerial wireless communication for the Internet of Vehicles (IoVs) [23–25], which function as dynamic IoT nodes. IoVs require high-speed, reliable, low-latency, and stable wireless links [26] to exchange perception data [27] critical for safe self-driving [28].

Device-to-device (D2D) communication is expected to be indispensable in future WCNs for enabling high-speed connectivity, offloading data traffic, and optimizing the capacity of large-scale networks [29]. IVs can leverage vehicle-to-vehicle (V2V) communication to establish vehicular networks [30] and enhance safety features such as collision avoidance for autonomous driving [31]. Additionally, IVs can utilize vehicle-to-everything (V2X) communication for applications related to commercial use, road safety, and traffic efficiency [32]. In this context, drones play a crucial role due to their high mobility and ability to disseminate urgent information swiftly among vehicles. They can also provide aerial LoS communication links for IoVs in obstructed, densely populated urban environments [33]. Given the inherent characteristics of future WCNs that often rely on LoS channels, integrating UAVs with RISs is essential for achieving more robust and efficient wireless communication [34,35].

UAV-enabled, RIS-assisted communication is generally formulated as a multi-objective optimization problem with the following joint objectives [34]:

1. Power transmission for propagating communication signals and/or power allocation to multiple UEs, where the power transmit is implemented by the terrestrial BS or UAV as the airborne BS.
2. UAV energy-efficient path planning and trajectory optimization, taking into account communication performance and quality of service (QoS) requirements.
3. Design of BS beamforming (BS-BF) and RIS phase shift (RIS-PhSh) to enhance signal quality and communication reliability.

In this context, with a focus on the integration of UAV-RIS in dense urban environments and wireless vehicular networks, our literature review identifies several research gaps, categorized into five main areas of study:

1. A new paradigm is required for channel modeling in the context of UAV-enabled, RIS-assisted communication for IoT sensors and particularly for mobile IoVs in dense urban environments. Notably, the channel modeling is based on localization-aware communication, which is promising due to the intrinsic characteristics of future B5G and 6G WCNs with mmWave signals, particularly considering dynamic IoTs and IoVs networks for vehicular communications.

2. Kinodynamic-aware, obstacle-free trajectory optimization for crash avoidance and navigation in dense urban areas has been largely overlooked in the wireless communication community and requires further exploration.
3. The emergence and applications of RIS-equipped UAVs (RISeUAVs).
4. A new perspective of optimization algorithms regarding the impact of trajectory optimization with LoS links.
5. Investigating integrated localization and communication within UAV-enabled, RIS-assisted communication systems for IoVs.

In the following Section, we present a taxonomy of UAV-enabled, RIS-assisted communication studies, which structured around the five main areas above.

2. Organization and Contribution of the Paper

The topics examined in this paper are illustrated in Figure 1. A significant portion of related research focuses on UAV placement, either for hovering or determining optimal navigation paths/trajectories. The UAV's navigation should be designed to maximize communication performance and associated metrics. Consequently, in addition to UAV navigation, key research areas include communication channel modeling, RIS phase shift design, and beamforming for MIMO base stations. To achieve these objectives, optimization problems are formulated based on defined goals and subject to physical constraints. Additional constraints often relate to communication metrics such as QoS, minimum achievable rate, signal-to-noise ratio (SNR), cognitive radio capabilities, multiple-access coding, or user-specific requirements.

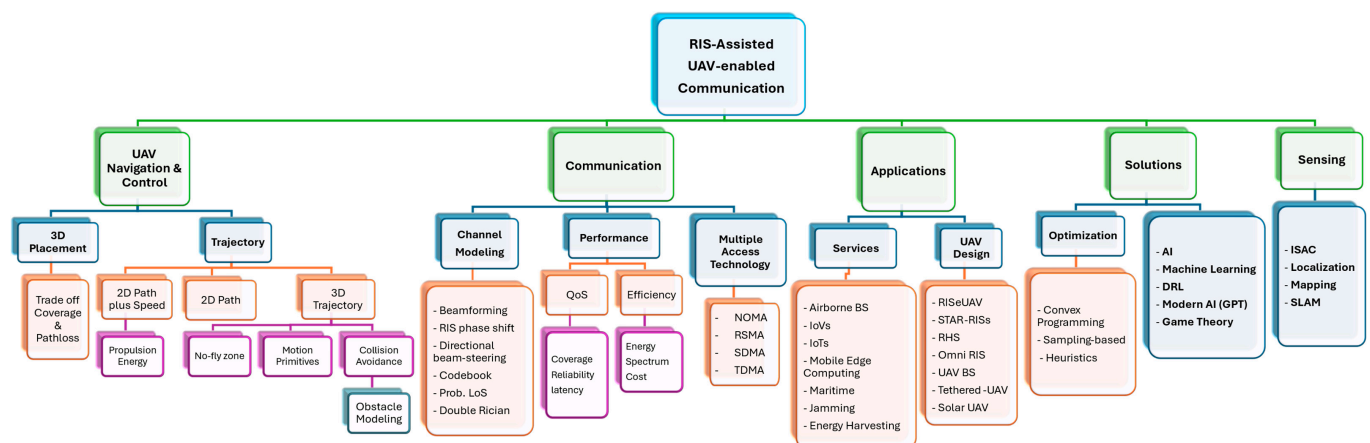


Figure 1. Organization of the paper based on the taxonomy of the UAV-enabled, RIS-assisted communication into quintuple studied and topics.

Minimizing energy consumption, both for UAV propulsion and transmitted communication power, is a recurring objective to enhance energy efficiency, sustainability, and cost-effectiveness. Since UAVs have limited onboard energy storage due to battery constraints, reducing propulsion energy consumption is crucial for extending UAV endurance during communication missions. This paper categorizes the research into five major sections, which are briefly explained following Remark 1.

Remark 1. The central focus of this paper is on “RIS-assisted UAV navigation in future wireless networks for supporting IoT and IoV communication in dense urban areas of smart cities.” This is an emerging research area, particularly relevant in the context of autonomous intelligent vehicles that require sensor fusion, incorporating visual data, V2X communication, and mmWave localization for safe and secure autonomous driving. This paper is structured from this viewpoint and adopts a tutorial approach, summarizing key concepts and methodologies found in the literature. Notably, related works often address multiple aspects of this taxonomy, as we summarize in a Table, which maps the contributions of various research papers to the following five main areas.

1. Section 3 delves into *communication and channel modeling*, examining key aspects such as channel modeling, network performance, and modern multiple-access communication technologies. UAV deployment and navigation are considered as strategies for augmenting existing terrestrial cellular networks or delivering on-demand or remote coverage. In these scenarios, the channel model, path loss, and phase shift significantly influence the performance of RIS-assisted, UAV-enabled communication and must be integrated into the optimization framework. Unlike conventional ground-to-ground (G2G) channels, air-to-ground (A2G) channel modeling presents unique challenges and characteristics [29]. Therefore, this Section reviews channel modeling specific to UAV-enabled, RIS-assisted communication, highlighting its distinct aspects. Additionally, network performance indices, such as QoS, throughput, coverage, and weighted sum rate (WSR), are mathematically formulated and incorporated into the optimization problem. Recent advancements in multiple-access communication technologies for cognitive networks are also explored, showcasing their ability to enhance spectrum and energy efficiency, thereby improving overall network performance and connectivity. Moreover, relevant works consider the coding and decoding order of these multiple-access techniques as part of the optimization problem.
2. Section 4 explores *UAV navigation and control*, a topic of significant importance from two critical perspectives. Firstly, energy efficiency is a major concern; UAV navigation, including localization, path planning, path tracking, and trajectory design for generating motion primitives, affects the UAV's energy input and consequently its propulsion energy consumption. Therefore, studies should incorporate UAV navigation considerations to enhance energy efficiency. Secondly, from a practical standpoint, generated paths and trajectories must be both dynamically and kinematically (kinodynamically) feasible. This means that UAV motion must adhere to kinematic constraints such as speed, acceleration, and nonholonomic limitations (e.g., rate of heading change). Moreover, in dense urban environments where paths are often obstructed and maintaining LoS links is crucial, the trajectory should also account for the UAV's rotational dynamics. This practical aspect is frequently overlooked in the current literature within the wireless communication field.
3. *Applications of UAVs and RISs in wireless networks* are discussed in Section 5, where various services involving UAV deployment in communication networks, UAV design, and RIS technologies are reviewed. Each application, whether it involves UAVs as airborne BS, IoT sensor data collection, support for V2X communication in IoVs, maritime communication, etc., comes with its own unique environment and network characteristics. These differences necessitate distinct objectives and constraints in the optimization problems and significantly influence UAV navigation and control models. Additionally, depending on the application or available resources, the aerial package, comprising the UAV and communication module, can vary in design. For example, UAVs may be deployed as active aerial base stations (UAV-BS) or equipped with RIS (RISeUAV) to enhance energy efficiency and reduce system complexity.
4. *Solutions* to the optimization problems are discussed in Section 6, covering convex programming, machine (deep) learning, and game theory approaches. Convex programming is a widely used solution for optimization problems. However, optimization challenges in UAV navigation, such as joint energy-efficient trajectory design and communication performance, are typically highly nonlinear and non-convex. As a result, alternating optimization (AO) techniques, including sequential convex programming (SCP) and successive convex approximation (SCA), are employed after applying convex relaxation methods. These techniques provide computationally tractable solutions, albeit at the cost of suboptimality [36]. Additionally, with recent advancements in artificial intelligence (AI), deep learning, and particularly deep reinforcement learning (DRL) agents, these methods have garnered significant attention.
5. Section 7 explores *integrated localization and communication*. The radio frequency sensor (radar) has been employed for object detection and tracking. In future WCNs,

mmWave communication and beamforming are expected to enable higher angular resolution, leading to more accurate centimeter-level localization. As a result, mmWave localization technology has garnered significant attention for autonomous unmanned vehicles [37]. Additionally, RISs are increasingly valuable for creating smart radio environments, where the propagation and scattering of communication signals can be controlled, reducing multipath interference. Furthermore, localization-assisted communication is instrumental in transmitting and propagating directional communication signals, enhancing the efficiency and effectiveness of communication. The integration of sensing (localization) and communication (ISAC) serves as a powerful technique to optimize resource utilization, network capacity, and spectrum efficiency.

3. Communication and Channel Modeling

In this section, we present a brief study on channel modeling and the impact of environment and RIS-assisted communication on channel performance. Through channel modeling, we highlight critical points that should be considered in the UAV navigation for UAV-enabled, RIS-assisted communication.

3.1. Channel Modeling

The received signal at the UE is affected by the channel gain between the transmitter and that point as the following:

$$y = \sqrt{P_T} g_{LoS}^{TR} s + w, \quad (1)$$

where y denotes the received signal, P_T denotes the transmitted power, g_{LoS}^{TR} denotes the LoS channel gain, s denotes the coded signal, and $w \approx \mathcal{CN}(0, \sigma^2)$ is a complex Gaussian additive noise with zero mean and covariance σ^2 for modeling scattering components. The channel gain may be adversely affected by the environment, resulting in NLoS multipath signals due to scattering and consequence random reflections, diffractions, and refractions, see Figure 2.

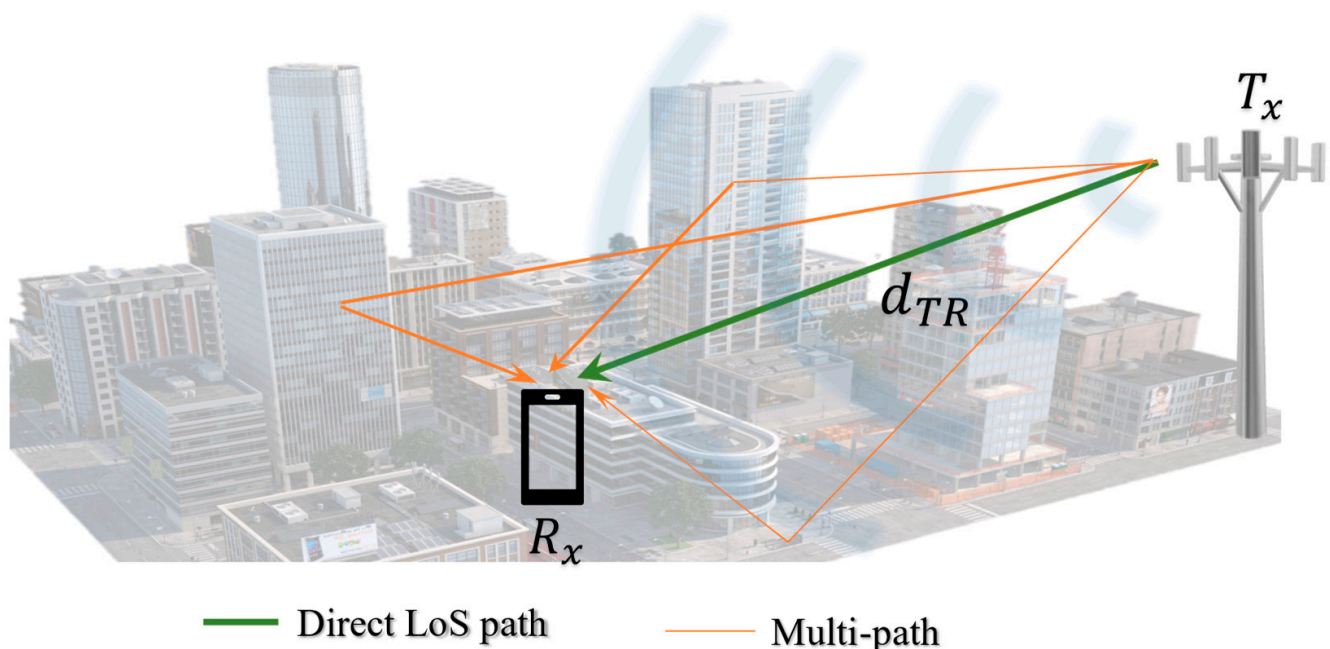


Figure 2. Illustration of direct LoS path and multi-path.

In future networks, the LoS signals are assumed to be dominant compared to NLoS multipath signals that are negligible based on real-world experimental [38]. The channel

(power) gain is determined by the free-space LoS path loss affected by the distance between a transmitter and receiver as follows:

$$\mathcal{G}_{LoS}^{TR} \approx \frac{\lambda}{4\pi} \sqrt{G_T} \sqrt{G_R} \frac{e^{-\frac{2\pi}{\lambda} d^{TR}}}{\underbrace{\|p^{Tx} - p^{Rx}\|_2}_{d^{TR}}} \quad (2)$$

where $\mathcal{G}_{LoS}^{TR} \in \mathbb{C}$ denotes the LoS channel gain between the transmitter T_x and receiver R_x ; G_T and G_R denote the transmitter and receiver antenna gain, respectively; $d^{TR} = \|p^{Tx} - p^{Rx}\|_2$ presents the distance between the transmitter and the receiver, with $p^{Tx} = [x^{Tx} \ y^{Tx} \ z^{Tx}]^T \in \mathbb{R}^3$ and $p^{Rx} = [x^{Rx} \ y^{Rx} \ z^{Rx}]^T \in \mathbb{R}^3$ denoting the 3D position (i.e., Cartesian coordinates) of the transmitter and receiver, respectively; superscript $\top$ denotes matrix transpose, and $\|\cdot\|_2$ presents Euclidean norm; ϕ^{TR} denotes the induced phase shift, and λ is the carrier wavelength.

Remark 2. The communication performance in B5G and 6G is affected by LoS paths where the channel gain of LoS paths is affected by the distance. Also, the LoS path may be randomly blocked in the case of mobile IoVs in urban environments. In UAV-enabled, RIS-assisted communication, UAVs may be deployed as airborne BSs (UAV-BS) to provide aerial LoS paths through RISs mounted on buildings. Alternatively, UAVs may be equipped with RISs as the RIS-equipped UAV (RISeUAV) to cover blind spots and provide extra (indirect) cascaded LoS (CLoS) paths for IoVs, see Figure 3.

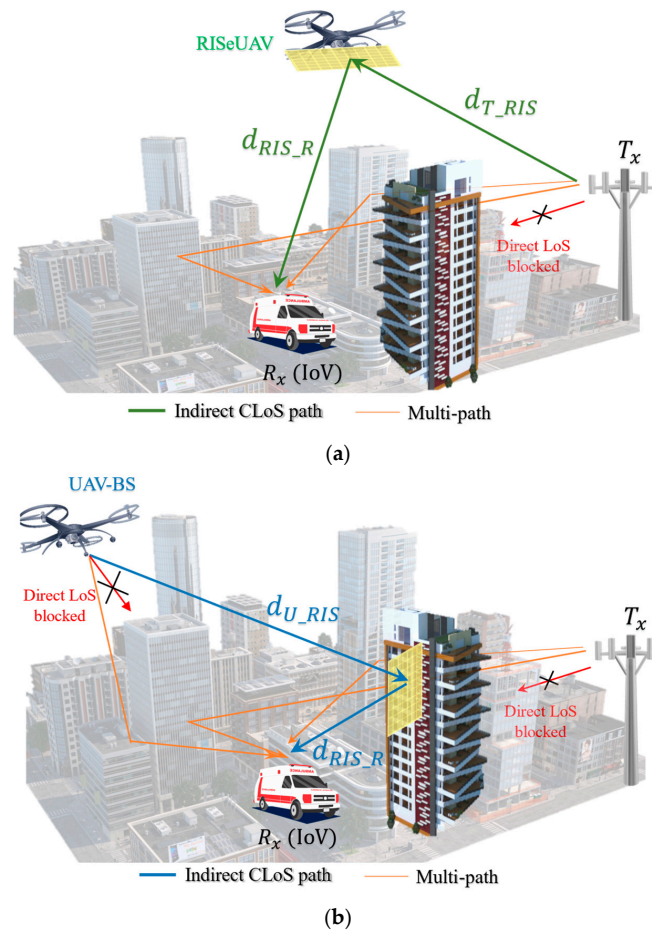


Figure 3. UAV-enabled, RIS-assisted communication: (a) RISeUAV with a UPA of the RIS aligned in the XY plane facing the ground; (b) UAV-BS as an active aerial (airborne) BS.

3.1.1. Deployment of UAV-BS and RISeUAV

After the deployment of a UAV equipped with an aerial RIS as the RISeUAV, the channel modeling is obtained as follows. The same methodology can be applied to the RIS-assisted UAV-BS.

The channel gain of the cascaded $T_x - \text{RIS} - R_x$ LoS path for the RISeUAV, in Figure 3a, is obtained as [39,40]:

$$\mathcal{G}_{\text{CLoS}}^{\text{TR}} \approx \frac{\lambda}{4\pi} \mathcal{G}_{\text{LoS}}^{\text{RIS-R}} \Psi^{\text{RIS}} \mathcal{G}_{\text{LoS}}^{\text{T-RIS}}, \quad (3)$$

where the notations of (3) is presented in the following: $\dagger$ denotes conjugate (Hermitian) transpose; $\mathcal{G}_{\text{CLoS}}^{\text{TR}} \in \mathbb{C}$ denotes the channel gain of the indirect CLoS link between transmitter and receiver through RIS (i.e., the $T_x - \text{RIS} - R_x$ link); $\Psi^{\text{RIS}} \in \mathbb{C}^{M \times M}$ denotes the RIS phase shift matrix given as follows:

$$\Psi^{\text{RIS}} = \text{diag}\{e^{j\psi_1^{\text{RIS}}}, e^{j\psi_2^{\text{RIS}}}, \dots, e^{j\psi_m^{\text{RIS}}}, \dots, e^{j\psi_M^{\text{RIS}}}\} \in \mathbb{C}^{M \times M}, \quad (4)$$

where ψ_m^{RIS} denotes the phase shift of the m^{th} RIS element, and M denotes the number of RIS reflector elements; $\mathcal{G}_{\text{LoS}}^{\text{T-RIS}} \in \mathbb{C}^M$ denotes the LoS channel gain of the $T_x - \text{RIS}$ link given as follows:

$$\mathcal{G}_{\text{LoS}}^{\text{T-RIS}} \approx \frac{\lambda}{4\pi} \frac{\sqrt{G_T}}{\underbrace{\|p^{T_x} - p^{\text{RIS}}\|_2}_{d^{\text{T-RIS}}}} \left[e^{-j\phi_{(m=1)}^{\text{T-RIS}}}, \dots, e^{-j\phi_m^{\text{T-RIS}}}, \dots, e^{-j\phi_M^{\text{T-RIS}}} \right]^\top \quad (5)$$

where $p^{\text{RIS}} = [x^{\text{RIS}} \ y^{\text{RIS}} \ z^{\text{RIS}}]^\top \in \mathbb{R}^3$ denotes the 3D position of the RIS; $\phi_m^{\text{T-RIS}} = \frac{2\pi}{\lambda} d_m^{\text{T-RIS}}$ indicates the imposed phase shift in the associated incident path (i.e., the path between the transmitter and m^{th} RIS element) where $d_m^{\text{T-RIS}} = \|p^{T_x} - p_m^{\text{RIS}}\|_2$; and $p_m^{\text{RIS}} = [x_m^{\text{RIS}} \ y_m^{\text{RIS}} \ z_m^{\text{RIS}}]^\top \in \mathbb{R}^3$ denotes the Cartesian coordinates of the m^{th} RIS element. $p_m^{\text{RIS}}, \forall m \in M$ can be obtained based on the RIS position and its spacing among its elements [41], see Figure 4 and Appendix A. $\mathcal{G}_{\text{LoS}}^{\text{RIS-R}} \in \mathbb{C}^M$ denotes the LoS channel gain of the $\text{RIS} - R_x$ link given as follows:

$$\mathcal{G}_{\text{LoS}}^{\text{RIS-R}} \approx \frac{\lambda}{4\pi} \frac{\sqrt{G_R}}{\underbrace{\|p^{\text{RIS}} - p^{R_x}\|_2}_{d^{\text{RIS-R}}}} \left[e^{-j\phi_{(m=1)}^{\text{RIS-R}}}, \dots, e^{-j\phi_m^{\text{RIS-R}}}, \dots, e^{-j\phi_M^{\text{RIS-R}}} \right]^\top \quad (6)$$

where $\phi_m^{\text{RIS-R}} = \frac{2\pi}{\lambda} d_m^{\text{RIS-R}}$ indicates the imposed phase shift in the associated reflection path (i.e., the path between m^{th} RIS element and the receiver), where $d_m^{\text{RIS-R}} = \|p_m^{\text{RIS}} - p^{R_x}\|_2$. Therefore, the equivalent channel gain of the indirect $T_x - \text{RIS} - R_x$ CLoS link can be obtained as follows:

$$\mathcal{G}_{\text{CLoS}}^{\text{TR}} \approx \left(\frac{\lambda}{4\pi} \right)^2 \sqrt{G_T} \sqrt{G_R} \frac{\sum_{m=1}^M e^{-(\psi_m^{\text{RIS}} + \phi_m^{\text{T-RIS}} - \phi_m^{\text{RIS-R}})}}{\underbrace{\|p^{T_x} - p^{\text{RIS}}\|_2}_{d^{\text{T-RIS}}} \underbrace{\|p^{\text{RIS}} - p^{R_x}\|_2}_{d^{\text{RIS-R}}}} \quad (7)$$

By assigning the phase shift for the m^{th} RIS element as follows:

$$\psi_m^{\text{RIS}} = \phi_m^{\text{RIS-R}} - \phi_m^{\text{T-RIS}}, \quad (8)$$

all reflected signals by the RIS elements arrive at the receiver with the same phase shift so they are inherently accumulated in the receiver. Therefore, the indirect CLoS is obtained as follows:

$$\mathcal{G}_{\text{CLoS}}^{\text{TR}} \approx \left(\frac{\lambda}{4\pi} \right)^2 \sqrt{G_T} \sqrt{G_R} \underbrace{\|p^{\text{Tx}} - p^{\text{RIS}}\|_2}_{d^{\text{T_RIS}}} \underbrace{\|p^{\text{RIS}} - p^{\text{Rx}}\|_2}_{d^{\text{RIS_R}}} \quad (9)$$

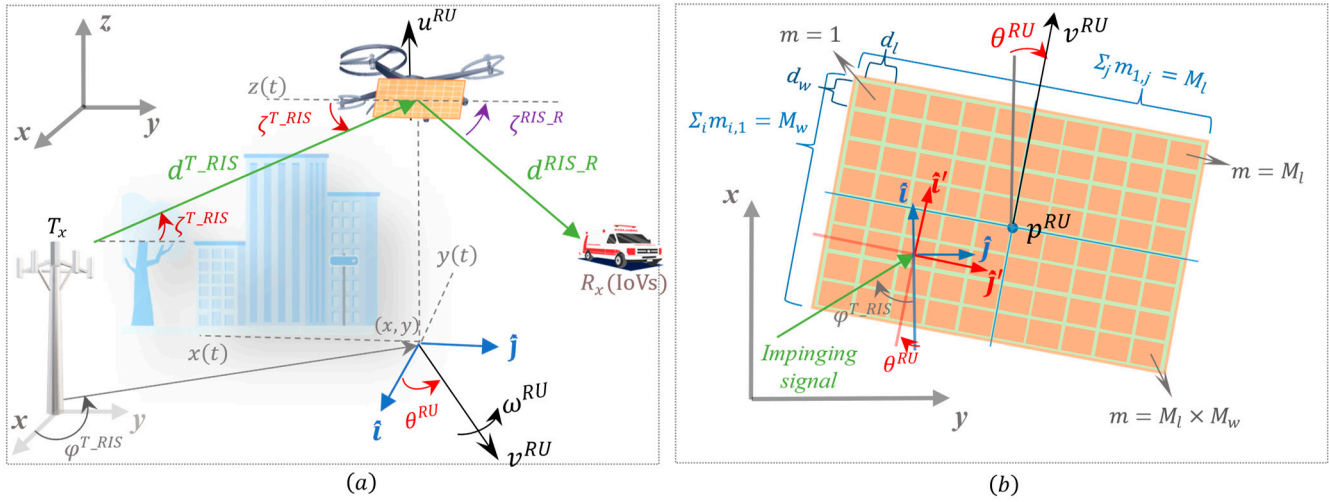


Figure 4. Schematic of RISEUAV-assisted communication for channel modeling: (a) geometry of system in 3D coordinates; (b) UPA of the RIS in XY plane; v^{RU} and u^{RU} denote UAV's horizontal and vertical linear velocities, respectively; ω^{RU} denotes the UAV's horizontal rotational velocity and θ^{RU} denotes the UAV heading (angle) with respect to the X-axis. The UAV motion is studied in Section 4.

Finally, the achievable rate is obtained as follows:

$$\mathcal{R} = \log_2 \left(1 + \frac{P_T |\mathcal{G}_{\text{LoS}}^{\text{TR}} + \mathcal{G}_{\text{CLoS}}^{\text{TR}}|^2}{\sigma^2} \right) \quad (10)$$

which is a concave function.

There are critical points given from the equivalent channel gain of the CLoS link which are highlighted in the following remarks.

Remark 3. In the channel gain of the CLoS link, the product of two Euclidean norms appears in the denominator, which makes the channel gain formula nonconvex, notably, when the optimization problem is the RISEUAV trajectory design. Notice that in (9), the RIS position p^{RIS} is determined by the UAV position. Handling this issue is the contribution of research works that are studied in the optimization solution Section (Section 6).

Remark 4. Although the product of the two Euclidean distances associated with the incident path to and reflection path from the RIS reduces the channel gain, the accumulated signals reflected by the RIS elements significantly enhance the channel gain, making it M times stronger, which compensates for the challenges of CLoS links. This notable improvement is achieved by using RIS without consuming energy or requiring complex processing. Since the phase shifts of RIS elements are passive, RIS presents a promising, sustainable solution to enhance the performance of future WCNs. However, the RISEUAV may need to fly at lower altitudes to increase channel performance, which introduces potential obstacles such as skyscrapers and urban canyons. Therefore, the trajectory optimization method must account for an obstacle-free path to avoid collisions.

Remark 5. The rationale behind the phase shift design in (8) is that perfect channel state information (CSI) is available, which is not practical in UAV-enabled communication, particularly for mobile IoVs. Therefore, the possible cost-effective solution is directional phase shift/beamforming,

considering that the localization information of the UAV-BS, RISEUAV, and UEs are available [42]. It is not an unwarranted assumption as the mmWave localization has been studied in the literature, which is discussed as the integrated localization and communication in Section 7.

3.1.2. Position-Aware Directional Beam-Steering

Based on Remark 5, directional beam-steering can be implemented to improve the channel gain and communication performance. Assuming the localization-aware communication and geometry-based directional phase shift, the channel gain can be obtained in the angular domain as follows:

$$\mathcal{G}_{CLoS}^{TR} \approx \left(\frac{\lambda}{4\pi}\right)^2 \sqrt{G_T} \sqrt{G_R} \frac{\overbrace{\left(\mathbf{a}_{inc}^{RIS^\dagger}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) \Psi^{RIS} \mathbf{a}_{ref}^{RIS}(\varphi^{RIS_{-}R}, \zeta^{RIS_{-}R})\right)}^{RIS}}{d_{T_{-}RIS} d_{RIS_{-}R}} e^{-(\phi^{inc} + \phi^{ref})} \quad (11)$$

$$\text{with } \begin{cases} \phi^{inc} = \frac{2\pi}{\lambda} \mathbf{w}^\top(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS})(p^{RIS} - p^{Tx}) \\ \phi^{ref} = \frac{2\pi}{\lambda} \mathbf{w}^\top(\varphi^{RIS_{-}R}, \zeta^{RIS_{-}R})(p^{Rx} - p^{RIS}) \end{cases}$$

where $\varphi^{T_{-}RIS}$ and $\zeta^{T_{-}RIS}$ denote azimuth and elevation angles of the *incident (inc)* path, respectively; $\varphi^{RIS_{-}R}$ and $\zeta^{RIS_{-}R}$ denote azimuth and elevation angles of the *reflection (ref)* path, respectively, which are given based on the geometry information, see Figure 4 and Appendix B; $\mathbf{a}_{inc}^{RIS_n}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) \in \mathbb{C}^{M \times 1}$ denotes the RIS array response (wave) vector to the incident signal from the transmitter to the RIS and is obtained as follows:

$$\mathbf{a}_{inc}^{RIS}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) = \left[e^{-j\phi_{(m=1)}^{inc}}, \dots, e^{-j\phi_m^{inc}}, \dots, e^{-j\phi_M^{inc}} \right]^\top \quad (12)$$

where ϕ_m^{inc} denotes the imposed phase shift to the signal received at the m^{th} element of the RIS with respect to the center point of RIS given as follows:

$$\phi_m^{inc} = \frac{2\pi}{\lambda} \mathbf{w}^\top(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS})(p_m^{RIS} - p^{RIS}); \quad (13)$$

$\mathbf{w}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) \in \mathbb{R}^3$ denotes the normalized 3D wave vector of the $T_x - \text{RIS}$ path given as follows:

$$\mathbf{w}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) = \begin{bmatrix} \cos(\zeta^{T_{-}RIS}) \cos(\varphi^{T_{-}RIS}) \\ \cos(\zeta^{T_{-}RIS}) \sin(\varphi^{T_{-}RIS}) \\ \sin(\zeta^{T_{-}RIS}) \end{bmatrix}; \quad (14)$$

Similarly, the RIS array response to the reflected signal $\mathbf{a}_{ref}^{RIS_n}(\varphi^{RIS_{-}R}, \zeta^{RIS_{-}R})$ can be achieved using the azimuth and elevation angles of the reflection path. The RIS phase shift design is implemented based on Remark 6.

Remark 6. Directional beam-steering is achieved by controlling the phase shifts (i.e., UPA phased array) of the RIS. It has been proven that by discrete Fourier transform (DFT)-based codebook design, the directional beam-steering is achieved if the DFT weight vector is equivalent to the normalized array response vector (i.e., the normalized wave vector) [43]. Therefore, the normalized array response vectors of the BS-RIS path and RIS-UE (e.g., RIS-IoVs) path are used to control the phased array for BS-BF and RIS-PhSh, respectively, to realize directional beam steering.

In this light, the phase shift matrix can be designed as follows:

$$\Psi^{RIS} = \mathbf{a}_{inc}^{RIS}(\varphi^{T_{-}RIS}, \zeta^{T_{-}RIS}) \mathbf{a}_{ref}^{RIS^\dagger}(\varphi^{RIS_{-}R}, \zeta^{RIS_{-}R}). \quad (15)$$

After using (15), (11) is converted to (9). However, the advantage of using a geometry-based phase shift design (i.e., directional beam-steering) is that it can be effectively implemented through an appropriate codebook design, as highlighted in Remark 6. Additionally, this approach is well-suited for dynamic vehicular networks, where vehicles are in motion. Positioning-aware beam steering can be employed as an alternative to CSI, which is challenging to obtain in such dynamic environments.

3.1.3. MIMO Beamforming

By deploying massive mMIMO BSs in future 6G networks [44] stronger signals can be propagated for IoVs, see Figure 5. In multiple antennas, the precoding (weighting) vector corresponding to the unit power signal is equal to the unit response vector of the linear antenna arrays to maximize the SNR. Therefore, based on Remark 6, the channel gain of the cascaded LoS path BS-RIS-IoV is obtained as

$$\mathcal{G}_{CLoS}^{BSRISIV} \approx \overbrace{\mathcal{G}^{BSRU}(\varphi^{BSRU}, \zeta^{BSRU})}^{BSRU} \times \overbrace{\left(\mathbf{a}_{inc}^{RIS\dagger}(\varphi^{BSRU}, \zeta^{BSRU}) \Psi^{RIS} \mathbf{a}_{ref}^{RIS}(\varphi^{BSIV}, \zeta^{BSIV}) \right)}^{RIS} \times \overbrace{\mathcal{G}^{RUIV}(\varphi^{RUIV}, \zeta^{RUIV})}^{RUIV} \quad (16)$$

where N is the number of antennas in the mMIMO BS; superscripts “BSRU” and “RUIV” denote the BS-RISeUAV (RU) link (i.e., corresponding to the incident (*inc*) signal with respect to the RIS) and RU-IV link (i.e., corresponding to the reflection (*ref*) signal with respect to the IV), respectively. $\mathcal{G}^{BSRU}(\varphi^{BSRU}, \zeta^{BSRU}) \in \mathbb{C}$ denotes the channel gain of the BS-RU path given as follows:

$$\mathcal{G}^{BSRU}(\varphi^{BSRU}, \zeta^{BSRU}) = \sqrt{G_{BS}} \mathbf{a}^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) \times \Gamma^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) \times \mathcal{G}_{Ploss}^{BSRU}, \quad (17)$$

where $\mathbf{a}^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) \in \mathbb{C}^N$ denotes the array response (wave) vector of the mMIMO BS with respect to the RU position given as follows:

$$\mathbf{a}^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) = \left[e^{-j\phi_{(n=1)}^{BSRU}}, \dots, e^{-j\phi_n^{BSRU}}, \dots, e^{-j\phi_N^{BSRU}} \right]^\top, \quad (18)$$

$$\text{with } \phi_n^{BSRU} = \frac{2\pi}{\lambda} \mathbf{w}'^\top(\varphi^{BSRU}, \zeta^{BSRU}) (p_n^{BS} - p^{BS})$$

$$\text{and } \mathbf{w}'(\varphi^{BSRU}, \zeta^{BSRU}) = \begin{bmatrix} \sin(\zeta^{BSRU}) \cos(\varphi^{BSRU}) \\ \sin(\zeta^{BSRU}) \sin(\varphi^{BSRU}) \\ \sin(\zeta^{BSRU}) \end{bmatrix},$$

$p_n^{BS} = [x_n^{BS} \ y_n^{BS} \ z_n^{BS}]^\top \in \mathbb{R}^3$ denotes the Cartesian coordinates of the n^{th} antenna of the mMIMO BS, which can be obtained based on the BS position and its spacing among its elements, see Figure 5 and Appendix C. $\Gamma^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) \in \mathbb{C}^N$ denotes the BS phase array

$$\Gamma^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) = \frac{1}{\sqrt{N}} \left[e^{-j\gamma_{(n=1)}^{BSRU}}, \dots, e^{-j\gamma_n^{BSRU}}, \dots, e^{-j\gamma_N^{BSRU}} \right]^\top, \quad (19)$$

$\mathcal{G}_{Ploss}^{BSRU} \in \mathbb{C}$ denotes the path loss of the BS-RIS path:

$$\mathcal{G}_{Ploss}^{BSRU} = \frac{\lambda}{4\pi} \frac{e^{-j\phi^{BSRU}}}{\underbrace{\|p^{RU} - p^{BS}\|_2}_{d^{BSRU}}},$$

where

$$\phi^{BSRU} = \frac{2\pi}{\lambda} \mathbf{w}^T(\varphi^{BSRU}, \zeta^{BSRU}) (p^{RU} - p^{BS});$$

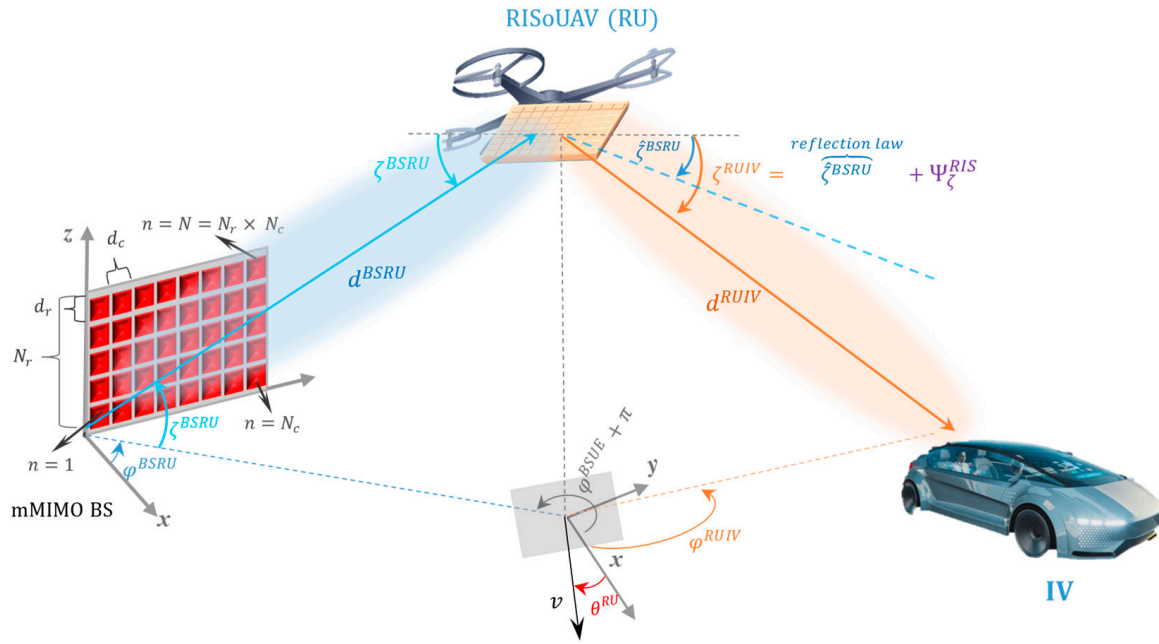


Figure 5. Schematic of UAV-enabled, RIS-assisted wireless communication for intelligent vehicles (IVs) in IoVs with mMIMO BSs. Notice that, for the sake of illustration, the sizes of the mMIMO BS and RISoUAV are exaggerated compared with the distances.

The array response vectors of the RIS for incident and reflected paths are obtained as presented earlier in (12)–(15). After applying the directional beam steering method for BS-beamforming (i.e., $\Gamma^{BS}(\varphi^{BSRU}, \zeta^{BSRU}) = \mathbf{a}^{BS*}(\varphi^{BSRU}, \zeta^{BSRU})$), and RIS phase shift design in (15), the channel gain can be approximated as follows:

$$\mathcal{G}_{CLoS}^{BSRISIV} \approx \underbrace{\left(\frac{\lambda}{4\pi}\right)^2 \sqrt{G_{BS}} \sqrt{G_{IV}} F_{inc}(\varphi^{BSRU}, \zeta^{BSRU}) F_{ref}(\varphi^{BSIV}, \zeta^{BSIV})}_{\text{Constant}} \frac{e^{-(\phi^{BSRU} + \phi^{RUIV})}}{\underbrace{\|p^{RU} - p^{BS}\|_2}_{d^{BSRU}} \underbrace{\|p^{IV} - p^{RU}\|_2}_{d^{RUIV}}} \quad (20)$$

where $F(\varphi, \zeta)$ denotes the cosine shape scan loss function [45] that is embedded in the channel modeling for modeling the mismatches between actual positions and estimated positions of mobile elements such as UAV or IVs:

$$F(\varphi, \zeta) = \begin{cases} \begin{pmatrix} \cos(\varphi_{LoS} - \varphi_{beam}) \\ \times \cos(\zeta_{LoS} - \zeta_{beam}) \end{pmatrix}, & \zeta \in [0, \frac{\pi}{2}], \varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}] ; \\ 0, & \zeta \in (\frac{\pi}{2}, \pi], \varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}] \end{cases} \quad (21)$$

with subscript “LoS” denoting the actual LoS path and the subscript “beam” denoting the propagated beam based on estimated positions.

Remark 7. By utilizing mMIMO beamforming technology and implementing directional beam-steering with narrow beams, higher angular resolution can be achieved, which enhances the accuracy of localization. This is a significant advantage of mmWave communication [46]. The channel model in (20) can be leveraged to enable cost-effective and computationally efficient localization. Localization is achieved by adjusting the beam-steering based on the estimated position and measuring the variation in signal strength. When the estimated position of the RISoUAV and IV approaches the actual position, the channel gains, and consequently the signal strength, are maximized. The signal strength can be used as a measure for accurate localization.

3.1.4. Obstructed LoS and Outage Probability

Channel modeling in UAV-enabled communication is mostly based on Rician fading and free path loss gain for compromising LoS links and scattering [47]. Rician distribution is a more realistic scenario since the UAV is in a high position in the sky and can create a direct LoS to UEs. However, since future RIS-assisted mmWave communication networks most likely will be limited to beamforming to LoS paths, deterministic LoS channel gain has been considered [48]. Also, probabilistic LoS is common in urban environments with buildings appearing as random obstacles. This issue was initially considered for coverage maximization of low-altitude aerial platforms (LAPs) [49]. Following this work, some papers have used the developed probabilistic LoS modeling for applications in urban environments [50,51]. The probability of LoS link between the UAV and ground UE in urban areas is obtained as follows:

$$\text{Prob}_{\text{LoS}} = \frac{1}{1 + a \exp \left(-b \left(\underbrace{\frac{180}{\pi} \arcsin \left(\frac{z^{\text{UAV}}}{d^{\text{UAV-UE}}} \right)}_{\zeta^{\text{UAV-UE}}} \right) - a \right)} \quad (22)$$

where a and b are constant values determined based on the environment; z^{UAV} denotes the UAV altitude; $d^{\text{UAV-UE}}$ denotes the Euclidean distance; $\zeta^{\text{UAV-UE}}$ denotes the elevation angle between UAV and UE. The outage probability is $1 - \text{Prob}_{\text{LoS}}$. Therefore, since the elevation angle in the deterministic LoS impact the channel, there is a tradeoff between elevation angle and distance that is considered in the RISeUAV (or UAV-BS) placement and path planning [52].

Further, more recently, the stochastic geometry of the LoS path has been considered in determining the LoS probability of the A2G links [53]. In the case of RISeUAV, it is assumed the LoS links are valid in the incident and reflection directions, thanks to the great mobility of the UAV [48]. Therefore, a deterministic CLoS channel model has been considered as given in (9). However, the channel model is affected by a variety of realistic physical disturbances affecting the channel conditions such as the UAV mobility [54] and weather conditions [55]. By incorporating the UAV's rotational aerodynamics, the array response of the RIS array in the RISeUAV has been modified following UAV fluctuations [54]. The environmental characteristics affect the small-scale and large-scale fading and potential spatial correlation in shadowing. Therefore, to realize a generic model, (9) is modified through a statistical study to develop a generic model [55], where the closed-form expressions for the outage probability and the ergodic capacity have been presented.

Also, the RISeUAV faces the double-Rician channel, due to the CLoS link that needs a new closed-form for modeling the outage probability [56]. The impact of the UAV position and the number of RIS reflective surface elements on the channel modeling and the performance of the communication network are investigated using Monte Carlo simulations [56]. Further, the variance of the double-Rician channel and a closed-form expression of the outage probability have been developed.

Realistic environment testing has characterized obstructed V2V LoS links, highlighting the impact of shadowing from other vehicles and self-shadowing on flat and sloped terrains [57]. Higher antenna placement (e.g., on vehicle roofs) can improve coverage in obstructed scenarios but may worsen coverage on sloped terrains due to self-shadowing. Deploying RISeUAV can effectively provide aerial indirect LoS links to enhance V2V communication.

3.1.5. Imperfect CSI

Normally, the communication strategies and trajectory optimization are based on the knowledge of channel coefficients such as full and partial CSI. The main issue with

UAV-enabled, RIS-assisted communication is imperfect CSI and channel estimation and beamforming in dynamic environments.

A UAV-BS-based multiple RIS-assisted communication with outdated and imperfect CSI has been studied in [58]. In the case of multiple RISs, the problem is RIS allocation to a given user. The statistical distributions of instantaneous SNR under imperfect CSIs for both direct and composite fading links were derived using Rician fading. Using the cumulative distribution function (CDF) of SNR, series-based expressions for average coverage probability (ACP) and average bit error rate over the UAV's flight were obtained.

To support ultra-reliable and low-latency communications (e.g., for IoTs and IoVs in B5G and 6G cellular networks [59]), short-packet transmissions are essential. In this light, a new framework for short-packet communications has been developed in [60], where an error probability bound for a given block length (BL) and coding rate have been derived. Followed by this, Agrawal, et al. investigated the application of RIS in energy harvesting in a UAV-RIS-IoT communication network with finite BL codes [61]. Using the derived statistics, the system's performance with finite block length codes was analyzed for average outage probability, block error rate, and goodput over the UAV's flight duration. The impact of factors such as imperfect CSI, number of RISs and reflecting elements, IoT locations, UAV altitude, and channel fading severity on UAV-RIS link performance was also investigated.

3.1.6. Codebook Design

Utilizing codebooks, with pre-designed codewords, is considered an effective and promising technique for handling channel estimation problems in dynamic environments including IoVs [62]. On top of IoVs' motion, the estimation of the end-to-end $T_x - \text{RIS} - \text{UAV} - R_x$ cascaded channel is more challenging due to the passive nature of RIS and limited onboard resources for the UAV's overhead controller. Therefore, the beamforming design is complicated in practice. However, beam training based on a predefined codebook can be used. Codebook design is based on offline beam training for beam alignment between two end nodes. However, an exhaustive search along the large conventional codebook, to find the optimal UPA phase shift matrix configuration, will cause extremely high system overhead and power consumption. Therefore, as pointed out in Remark 6, a positioning-based codebook with an affordable size has been utilized [63]. The proposed predefined codebook uses the estimated positions as initial points for determining the main reflecting pattern. Then, depending on the standard deviation of the localization error, sets of sampled points are determined for steering the beam. Also, optimization-based codebook design techniques have been considered utilizing the Karush–Kuhn–Tucker (KKT) condition to solve the problem with low-precision non-uniform codebooks [64] and the steepest-descent method (SDM) [65].

3.2. Communication Performance

Communication performance plays a crucial role in the optimization of UAV navigation and trajectory design, and it is integrated into the objective function and optimization problems. Key performance indicators for communication systems in UAV-based networks are QoS, secrecy, reliability, and efficiency in terms of energy and spectrum utilization. These factors ensure that the UAVs can maintain reliable, secure, and efficient communication links with UEs or other UAVs, thereby enabling seamless operation in dynamic environments.

3.2.1. Quality of Service (QoS)

QoS is a critical communication performance metric that dictates the overall user experience in terms of data rate, latency, and connection reliability. In UAV-based networks, particularly those utilizing UAVs, RIS, and UEs, the primary objective for QoS is often the maximization of the sum rate of the UAV-RIS-UEs LoS communication channels [40]. The sum rate optimization aims to achieve the highest possible throughput while adhering

to minimum achievable rate constraints for UEs, ensuring that each device obtains an adequate data rate for its operation.

In the context of UAVs, RISs can be deployed to enhance signal quality by intelligently reflecting and directing signals to specific UEs, thus improving the communication channel's performance. The UAV serves as a mobile base station or relay, and the RIS acts as an intelligent medium to adjust and optimize the propagation conditions between the UAV and UEs. This approach enables better control over the channel conditions and allows for more efficient utilization of the available spectrum, especially in complex environments with high interference or obstructions.

The objective function in such optimization problems typically incorporates factors such as the total system throughput, latency, and reliability. Constraints are imposed to ensure that the QoS requirements for UEs, such as minimum data rates or maximum latency, are met, see Table 1.

3.2.2. Secrecy, Security and Reliability

The secrecy and reliability indexes of UAV-enabled, RIS-assisted communication have been considered in several studies. These studies typically focus on wireless communication systems in the presence of eavesdroppers, with an emphasis on enhancing physical layer security through techniques such as beamforming, artificial noise injection, and signal rerouting. The secrecy performance of integrated satellite UAV-relayed RIS-assisted networks, with multiple vehicle eavesdroppers, has been investigated in [66]. In these systems, the UAV forwards the legitimate signal to the destination user through RIS, which ensures reliable transmission. The maximal ratio combining (MRC) eavesdropping technique is used to intercept the legitimate signal. Based on the proposed secrecy system model and the MRC eavesdropping scheme, the study analyzes the secrecy outage probability (SOP) across various SNRs, evaluating the effects of key parameters on the SOP. The theoretical analysis is validated through Monte Carlo (MC) simulations.

UAV trajectory optimization has been explored for robust and secure RIS-assisted communication in the presence of eavesdroppers [67,68]. The objective function of the joint trajectory and communication optimization problem aims to maximize the worst-case secrecy rate. For the RIS-UAV, it has been demonstrated that a well-planned flight path significantly enhances the security gain of physical layer security [69]. Moreover, the flexibility of the RIS-UAV is shown to be particularly effective in improving security when the transmission environment is favorable, such as higher transmit power or a weaker wiretap link. Additionally, the 3D positioning of the RIS-UAV and the partitioning of the RIS (i.e., dividing the RIS into several sections and controlling the phase shift of each section) have been optimized to increase the physical layer security for legitimate users while enhancing the impact of artificial noise on illegitimate users [70].

3.3. Efficiency with Multiple-Access Technologies and Cognitive Radio Communication

To meet the high-performance demands of future wireless networks, such as high capacity, low latency, and seamless connectivity for IoT sensors and IoVs, it is essential to optimize efficiency [71]. Alongside RISs and STAR-RISs, which enable the creation of smart radio environments, next-generation multiple-access technologies play a crucial role in maximizing connection density and enhancing spectral efficiency [72]. Cognitive radio communication allows smart devices to dynamically detect available channels within the same frequency band, improving spectrum utilization [73]. Common multiple-access technologies include non-orthogonal multiple access (NOMA), rate-splitting multiple access (RSMA), space division multiple access (SDMA), and time division multiple access (TDMA) [74].

3.3.1. Non-Orthogonal Multiple Access

Non-Orthogonal Multiple Access (NOMA) enables simultaneous communication with multiple users using the same physical resources (i.e., time, frequency, and coding) by

leveraging superposition coding at the transmitter and successive interference cancelation (SIC) at the receiver [75]. The error probability in RISEUAV-based NOMA systems has been investigated in [76,77], accounting for practical constraints such as hardware impairments (HWI) at the transceivers, inter-cell interference (ICI), and imperfect channel state information (CSI).

3.3.2. Rate-Splitting Multiple Access

RSMA has emerged as a generalized multiple-access technology that effectively manages interference in multi-user communication systems [78]. RSMA is particularly beneficial for UAV-enabled, RIS-assisted communication in multi-user IoV networks, enabling efficient operation within the same spectrum [79]. The optimal RSMA power allocation coefficients for UAV-RIS-IoV links are determined by minimizing the sum of the average outage probability (AOP) across all targeted vehicles.

Table 1. Summarizing list of the literature review.

Refs.	Navigation and Control	Application		Communication		Optimization Problem			
		Service	UAV Type RIS Tech.	Channel Model	MA Technique	OF	Formulation	Constraints	Optimizer
[39]	2D Path RIS-PhSh	Ground UE	UAV-BS	Rician channel	—	Max Ave. Ach. rate	Convex Programming	—	Convex solver CVX
[47]	2D Path RIS-PhSh	multi-cell multi-mobile-user	RIS-UAV	Rician channel	—	Min Ave. ergodic rate	SCA	BS power	Convex solver CVX
[48]	2D Path RIS-PhSh	Throughput maximization	RIS-UAV	Deterministic LoS	—	Max throughput	Iterative optimization	Speed, Power	Convex programming
[50]	3D path+speed RIS-PhSh	IoTs	UAV-BS	Probabilistic LoS	—	Max Ach. rate	Block coordinate descent- SCA	Velocity Acceleration	Interior-point method
[51]	2D Path+Speed RIS-PhSh	MTC	UAV-BS	Probabilistic LoS	—	EE	SCA	QoS Latency	Convex solver CVX
[52]	2D Path RIS-PhSh	Ground nodes	RIS-UAV	Probabilistic LoS	—	Max Ach. rate	MINP	Elevation angle	Interior point CVX solver
[69]	2D Path RIS-PhSh	Security	RIS-UAV	Free space loss	—	EE Ach. rate	MDP	Power	Gradient DRL-DDPG
[70]	2D Path RIS-PhSh	Physical layer security	RIS-UAV	Probabilistic LoS	—	Ergodic Secrecy	Non-convex	QoS, SINR	Convex solver CVX, PSO, SA
[75]	3D Placement, RIS-PhSh	Multiple UEs	RIS-UAV	LoS path loss	NOMA	Max Achi rate	SCA SDP	—	CVX solver, PSO
[77]	2D Path RIS-PhSh	Ground UE	RIS-UAV	Probabilistic LoS	NOMA	Ach. rate	Block coordinate descent	—	Riemannian gradient
[78]	3D Placement, RIS-PhSh	GUs	T-UAV	Rician ProbabilisticLoS	RSMA	WSDR	SCA	Min Rate and Power	AO and SCP
[80]	3D Placement, RIS-PhSh	Blocked GUs	UAV-BS	Rician fading	—	EE (BF) + SR	MINP	Min SR, Power	BCD: Adam+GA
[81]	3D Placement, RIS-PhSh	Security Anti-eavesdroppers anti-jamming	RIS-UAV	small-scale Rician fading	—	Min Max WCSR	MDP	Min DR, Power QoS	DRL-DQN-FFM
[82]	3D Placement, RIS-PhSh	Anti-eavesdroppers	RIS-UAV STAR-RIS	Rician fading	NOMA	Min SEE	SCA	Min DR,	DRL-DDQN
[83]	3D Placement, RIS-PhSh, EH	Maritime anti-jamming	RIS-UAV	LAP Probabilistic LoS	—	AR + AEH	Model-free	SNR and Power	DRL SoftMax DDPG
[84]	3D Placement, RIS-PhSh, Active BF	Static GUs	MAT UAV-BS	Rician channel model	NOMA	EE	SCA	Min DR and Power QoS	MRT GRP
[85]	3D Placement, RIS-PhSh, Active BF	Multiple UEs	Multi RIS-UAV	Rician channel	—	Max WSR	SCA SDP	Power	Interior point method
[86]	3D Placement, RIS-PhSh, Active BF	Multiple UEs	Swarm RIS-UAV	Rician channel Probabilistic LoS	—	Max WSR	AO Biconvex Programming	Power	Lagrangian dual method
[87]	2D path+speed RIS-PhSh	Video streaming	UAV-BS	Rician channel	—	QoS	AO SCA	QoS Time delay	P-BCD Taylor approx.
[88]	3D Trajectory	Ground target	UAV-BS	Probabilistic LoS	—	EE	Kinodynamic piece-wise approx.	UAV Dynamics	Gradient
[89]	3D Path+Speed RIS-PhSh	Ground UE	UAV-BS	Probabilistic LoS	—	EE Ach. rate	Model free	3D borders	DRL DDPG
[90]	2D Path RIS-PhSh	Ground vehicles	RIS-UAV	LoS path loss	NOMA	EE Ach. rate	MDP	Power	DRL DDPG
[91]	2D Path RIS-PhSh	Covering hotspots	RIS-UAV	Probabilistic LoS	—	Max Ach. rate	Energy-aware MAB	Energy	MAB
[92]	2D Path RIS-PhSh	IoT devices	UAV-BS	Rician channel	TDMA	Max sum-rate	SCA, Stackelberg game	Power Energy price	CVX solver BCD

Table 1. Cont.

Refs.	Navigation and Control	Application		Communication		Optimization Problem			
		Service	UAV Type RIS Tech.	Channel Model	MA Technique	OF	Formulation	Constraints	Optimizer
[93]	2D Path RIS-PhSh	Ground Vehicles	RISeUAV	free space path loss	—	Max Ach. rate	Mixed integer SCA	Power	Convex solver CVX
[94]	2D Path RIS-PhSh	V2V	RISeUAV	LoS path loss	—	Max Ach. rate	AO SCA	—	Convex solver CVX
[95]	Aerial backhauling	Traffic monitoring	LAP UAV-BS HAP RISeUAV	FSPL AR of ULA	—	EE, SNR	AO	FHRE	Weiszfeld
[96]	2D path IOS-PhSh	UAV Downlink Rate Enhancement	UAV-BS IOS	Rician Channel	—	MAX SNR	AO	—	Convex solver CVX
[97]	2D Path+Speed RIS-PhSh	Mobile Edge Computing	RISeUAV	Rician Channel	—	EE	SCA	CPU frequency	Dinkelbach
[98]	2D Path STAR-RIS-PhSh	Multiple Ground UEs	STAR-RIS	Rician Channel Rayleigh fading	—	Max SR	MDP	Min DR Power	DRRL
[99]	2D Path RIS-PhSh	MTC for IoTs	RISeUAV	Rician Channel	NOMA	Max Min Rate	SCA	Min SNR	Convex solver CVX
[100]	2D Path	IoVs	Multi UAV-BS	Rician Channel	—	EE Max Ach. rate	SCA	Min Ach. rate	Convex solver CVX
[101]	2D Path+Speed RIS-PhSh	IoVs	RISeUAV	LoS path loss	—	Max Ach. rate	MDP	Min Ach. rate	DRI DDPG
[102]	2D Path RIS-PhSh	Secrecy, Cognitive communication	RISeUAV	Rician fading	—	Max Ach. rate	MDP	Min Ach. rate	DRI DDPG, DCCN
[103]	2D Path RIS-PhSh	AoI IoTs	RISeUAV	Rician fading	—	Min AoI	Iterative BCD	Min Ach. rate	Convex solver CVX
[104]	2D Path+Speed RIS-PhSh	MEC IoTs	UAV-BS RISeUAV	Rician Channel	—	EE CPU allocation	Mixed nonconvex	CPU frequency	Double DQN
[105]	2D Path RIS-PhSh Power transmit	Secure MEC	UAV-BS IOS	Rician fading	—	EE Ach. rate	AO, SCA, LP, SDR	CPU frequency	CVX Branc and bound
[106]	2D Path RIS-PhSh	Improve QoS	UAV-BS IOS	Rician fading	TDMA	EE Ach. rate	Mixed integer non-convex SCA, BCD	Min Ach. rate	Convex solver CVX
[107]	2D Path RIS-PhSh Dynamic BS-BF	V2V IoVs	UAV-BS	—	—	EE Ach. rate	MDP	Power	DRL DDPG
[108]	2D Path RIS-PhSh	MEC	RISeUAV	Probabilistic LoS	—	EE	MDP	CPU frequency	DRL
[109]	2D Path RIS-PhSh	Covert communication	RISeUAV	Rician fading	—	EE Ach. rate	AO SCA	Power	Convex solver CVX
[110]	2D Path+Speed RIS-PhSh Power transmit	Aerial LoS links for UEs	RISeUAV	Rician fading	—	EE Ach. rate	AO SCA	Power Min Ach. rate	Convex DRL Heuristics (WOA)
[111]	2D Path RIS-PhSh BS-BF	Satellite-UAV- Terrestrial	RISeUAV	Rician fading	—	Ergodic rate	MDP	UAV Energy	DRL LSTM-DDQN
[112]	2D Path+Speed RIS-PhSh BS-BF	Multiple UEs	RISeUAV Active RIS	Rician fading	RSMA	EE Rate/Power	MDP	EE	Meta DRL SAC DDPG
[113]	2D Path RIS-PhSh	Maritime Anti-jamming	RISeUAV	Probabilistic LoS	—	EE	MDP	QoS for Maritime UE	DRL PPO
[114]	2D Path RIS-PhSh	Disaster management	UAV-BS STAR-RIS	Rician fading	NOMA	—	MDP	Min Ave.Rate	DRL PPO

4. UAV Navigation and Control

The channel model and communication performance indices of the UAV-aided communication depend on the UAV position in the 3D coordinates. Also, when it comes to multiuser scenarios or remote areas, the UAV path impacts communication performance. However, the onboard battery capacity is limited, and energy consumption for hovering and propulsion must be considered to obtain a minimum path with minimum energy consumption for the endurance of the navigation. Nevertheless, in the context of UAV-enabled communication support for mobile nodes, such as IoVs, the UAV speed should be taken into account in the path-planning technique. Moreover, in dense urban areas with buildings and skyscrapers appearing as obstacles, obstacle-free trajectory optimization for crash-avoidance navigation is necessary where motion constraints such as speed, acceleration, and nonholonomic must be considered [20,115]. The nonholonomic constraint, which is explained as the heading change limit of fixed-wing UAVs, has been considered in a few works in the wireless communication community, e.g., as curvature-constrained

trajectory [116]. Nevertheless, quadrotor drones possess rotational limitations that can be modeled as nonholonomic constraints. In this light, the UAV dynamic-aware kinematic (kinodynamic) motion model should be used [117], which is the main shortcoming in the wireless communication community.

Although these motion constraints with kinodynamic models have been extensively used in the robotic and control communities [118,119], UAV navigation in the context of wireless communication brings other objectives and constraints to the optimization problem which makes it computationally more complicated. Notably, for autonomous navigation, the trajectory optimization, including both channel performance and navigation indices, must be solved in real-time and without relying on external resources. We review relevant works and highlight the shortcomings from this critical perspective for dense urban areas.

4.1. UAV Placement

In most studies, UAVs are deployed to deliver wireless communication coverage for stationary or quasi-stationary UEs [120]. Examples are on-demand coverage when the terrestrial cellular network is not available or congested due to natural disasters or social events [121]. Further, a team of UAVs may be adopted to create an aerial cellular network [122], where RISs are deployed on uneven terrains to assist the UAV-BSs in eliminating coverage blind spots and extending coverage. In such situations, the 3D allocation and placement of the UAV(s) are important factors in enhancing the coverage and channel performance [75], e.g., there is a compromise between channel gain and coverage [49].

The placement of the RIS-assisted UAV-BS has been considered in [80] to provide wireless channels for blocked ground users. The UAV-BS placement problem is solved using the Adam optimizer, a gradient-based optimization, which is a popular optimization algorithm for solving stochastic optimization problems in machine (deep) learning [80]. The RISeUAV takes advantage of both UAVs and RISs simultaneously to improve future WCNs by reflecting signals in the sky. However, high-quality aerial LoS links, imperfect CSI in highly dynamic environments, etc., make the RISeUAV-assisted channel links more vulnerable to both adversarial eavesdropping and malicious jamming. In [81], a secure communication strategy for RISeUAV was proposed to maximize secrecy rates while meeting QoS requirements against eavesdroppers and jammers. The method jointly optimized transmit beamforming, artificial noise, RISeUAV placement, and RIS phase shift to handle mixed-attack scenarios. To solve the non-convex optimization problem in dynamic environments, a post-decision state deep Q-network (DQN) with Fourier feature mapping (DQN-FFM) was introduced for robust anti-attack transmission.

With STAR-RISs, the coverage of the RISs is extended to 360 degrees by simultaneous transmission and reflection [123]. RISeUAV with STAR-RIS can increase transmission security by reducing interference and controlling unfavorable environments for signal eavesdropping. Therefore, the placement of RISeUAV with STAR-RIS has been studied [82]. An iterative method was used for UAV placement in static scenarios, with fractional programming, SCA for power control, semidefinite relaxation for transmission/reflection design, and a search-based approach for STAR-RIS positioning. For mobile scenarios, a double DQN (DDQN) algorithm was employed to learn an online UAV path design policy from a long-term perspective.

Further, RISeUAV can be beneficial in a maritime communication system against jamming attacks [83]. The RISeUAV placement and RIS phase shift are determined using the DRL technique targeting energy efficiency, BS transmit power constraints, energy harvesting in simultaneous wireless information and power transfer (SWIPT), and QoS requirements for maritime UEs.

To obviate concerns regarding the limited onboard energy and payload, the application of a tethered UAV as an airborne BS has been proposed [78], where the tether provides a stable power supply and backhauling [124]. The limitation imposed by the limited length of the tether, particularly of cell-edge UEs, can be compensated by mounting RIS on the

surrounding buildings. The placement of the tethered UAV along with the RIS phase shift, is optimally determined in [78] to maximize the weighted sum data rate (WSDR) of ground users by using RSMA as communication technology.

The coverage area may be large and one UAV cannot cover all UEs; therefore, a network of UAVs can be employed [84]. In these situations, the hovering position of UAVs is determined by taking the inter-UAV distance as a constraint.

4.2. Path Planning and Trajectory Design

Based on the existing papers and practical considerations, we have classified the literature that has considered UAV navigation into three categories:

- (1) Two-dimensional path planning where UAV flies at a fixed altitude and the UAV waypoints are optimized through the path in a horizontal plane.
- (2) Two-dimensional path-plus-speed considering the UAV energy consumption for propulsion.
- (3) Three-dimensional trajectory optimization [125] while considering practical UAV navigation factors and motion constraints such as:
 - (i) Obstacle-free trajectory planning for crash avoidance navigation [126].
 - (ii) Considering kinematic motion constraints such as speed, acceleration and nonholonomic constraints [127,128].
 - (iii) Dynamic-aware motion primitives or input jerks of the trajectory to eliminate path tracking controller [129].

Remark 8. From the practical point of view, a path-tracking controller is required for path-planning optimization methods that determine waypoints of the UAV path rather than UAV motion inputs.

Remark 9. The difference in path planning (in the first group of works) with trajectory design is that, in the latter, time is an extra factor that is considered in the optimization problem. Therefore, in addition to the waypoint, the UAV speed along the path should be optimally determined (second group). Nevertheless, when it comes to obstacle-free UAV navigation in dense environments (e.g., for IoVs support in smart cities) kinodynamic-aware motion constraints and motion primitives should be determined to make the trajectory practically feasible (third group).

Remark 10. In the majority of relevant works in the wireless communication community, obstacle-free navigation considering motion constraints has been overlooked [36,40,130], which makes it impractical from the robotic navigation perspective in practice. Additionally, in some works in the wireless communication community, the path planning is called trajectory, which is misleading from the robotic and UAV motion perspectives.

4.2.1. Two-Dimensional Path Planning

Basically, in the 2D path planning, the optimization variables are the UAV-BS or RISeUAV waypoints $p^{RU} = [x[k] \ y[k] \ z_{\text{Fixed}}]^T$ at consecutive time samples (slots) to maximize the achievable rate and throughput of UEs. Therefore, based on the achievable rate formula in (9) and (10), the optimization problem is defined as follows:

$\mathcal{P}1 :$

$$\text{OF1. } \max_{p^{RU}} \log_2 \left(1 + \underbrace{\frac{\|p^{RU} - p^{BS}\|_2}{d^{BSRU}}}_{d^{BSRU}} \underbrace{\frac{\|p^{RU} - p^{UE}\|_2}{d^{RUUE}}}_{d^{RUUE}} \right)$$

s. t.

$$\text{C1}_1. \quad \|p^{RU}[k+1] - p^{RU}[k]\|_2^2 < D$$

where $P^{RU} = \{p^{RU}[k], \forall k \in \{1 \sim K\}\}$ denotes the path including waypoints corresponding to discretized samples times from 1 to K ; $D = \delta V_{max}$ denotes the maximum distance the UAV can fly at each sample time based on its maximum linear speed V_{max} .

Based on Remark 3, the objective function OF1 is nonconvex, which is studied in Section 6. Constraint C1₁ imposes the speed constraints. However, as can be seen in $\mathcal{P}1$, the UAV motion constraints, including velocity/acceleration/nonholonomic constraints, as well as obstacle-free navigation, have not been considered. Moreover, an extra control loop is required for path tracking. Therefore, the produced path may not be feasible for practical applications, unless it is for long obstacle-free paths.

The 2D trajectory optimization of the RISEUAV has been studied in [48] by considering power transmit and RIS phase shift as joint objectives. The joint optimization problem is solved with iterative optimization and decomposing the OF into three optimization problems. Only UAV speed has been considered as a constraint, and motion constraints and obstacle-free navigation have not been considered. The finite flying time has been discretized into N ; some time slots and the waypoints of the 2D trajectory are obtained for each sample time that maximizes the objective function, which is the throughput of UEs. It is almost the case in many relevant works in 2D path planning, see Table 1.

4.2.2. Two-Dimensional Path-Plus-Speed

Along with the UAV 2D path, the UAV speed is optimized based on the UAV energy consumption for propulsion which consumes a major part of the overhead energy. The propulsion energy consumption of a rotary-wing UAV is obtained based on the UAV speed [131]:

$$\mathcal{E}^{UAV} = \delta \left(P_0 \left(1 + \frac{3\|v\|^2}{r^2\omega^2} \right) + \frac{d_0\rho s A_r}{2} \|v\|^3 + P_i \left(\sqrt{1 + \frac{\|v\|^4}{4v_0}} - \frac{\|v\|^2}{2v_0} \right)^{\frac{1}{2}} \right), \quad (23)$$

where $\mathcal{E}^{UAV}$ denotes propulsion energy consumption at sample time δ as a function of the UAV linear speed v . Physical parameters of the quadrotor impacting the energy consumption are given in Table 2. The propulsion energy consumption formula given in (24) has been used as the objective function to optimize the UAV speed [87], and some other works that studied 2D path-plus-speed (or acceleration [132]), are given in Table 1.

Table 2. Rotary-wing physical parameters affect the propulsion energy consumption.

Symbol	Parameter	Unit
v_0	The mean rotor velocity in hovering	m/s
P_0	Blade profile power in hovering	W
P_i	Induced power in hovering	W
ω	Rotor angular velocity	Radian/seconds
d_0	Fuselage drag ratio	—
A_r	Rotor disk area	m ²
r	Rotor radius	m
s	Rotor solidity	m ³
ρ	Air density	Kg/m ²

However, there are several critical issues with this model. First, it imposes significant computational overhead due to the need to solve the optimization function, including the non-convex formulation in (24). Additionally, using this model for optimizing the UAV's 3D trajectory in dense urban areas with obstacles is computationally prohibitive and inefficient. In environments with mobile vehicles and IoVs, the UAV's speed must be adaptable to maintain LoS paths for vehicles, as a fixed speed is impractical. Furthermore,

when the UAV hovers, its speed and energy consumption are fixed and known. Therefore, this model may be useful for obstacle-free long-distance flight at fixed speed (optimal speed) but is not practical in dynamic urban areas with mobile UEs.

4.2.3. No-Fly Zone

The 2D path planning and the 2D path-plus-speed optimization techniques are suitable for stationary, quasi-stationary UEs in wide and non-dense areas. However, these methods are not effective for UAV navigation in dense urban areas and, particularly, for mobile UEs and IoVs. Notably, in the RISEUAV or ground RIS-assisted UAV-BS communication, the channel gain of the indirect CLoS link significantly increases when the UAV altitude is high. Therefore, the UAV may be required to fly at lower altitudes among the buildings to provide LoS links for IoVs. It makes the crash avoidance navigation critical which should be considered in the trajectory optimization function [126].

In some works, the concept of no-fly zones has been considered [130,133] that should not be violated by the UAV path. L. Wang, et al. [89] have considered 3D path-plus-speed optimization for RIS-assisted UAV communication for ground UEs. No-fly zones have been modeled as Min and Max boundaries of 3D Cartesian coordinates. However, when it comes to real-time trajectory optimization for autonomous navigation, the formulation is problematic because modeling obstacles is a mixed-integer programming [134] in the context of convex programming, which imposes computational burdens. Moreover, kinodynamic motion constraints have not been considered. To this end, the third group of works considered 3D trajectory optimization using the UAV kinematic equation of motion.

4.2.4. Three-Dimensional Trajectory

In the third group of works, which includes only a few studies, the UAV kinematic equation of motion has been used for trajectory design [135]. Also, obstacle-free navigation and motion constraints are modeled to make the trajectory feasible. Further, the motion primitives of input jerks generated by the trajectory optimization problem can be used for UAV control without an extra controller required for path tracking. In this section, we consider factors affecting the UAV trajectory design such as obstacle-free path planning for crash avoidance as well as motion constraints to make the generated trajectory feasible in practice.

To the best of our knowledge, [88] is a rare work in wireless communication that has considered dynamic-aware 3D trajectory optimization. In this work, the conversion efficiency of electric motors, associated with quadrotors, has been considered in the trajectory optimization problem. For this reason, the energy consumption of the quadrotor for hovering and flight is obtained based on the power consumption of the rotors that is developed by using electromechanical formulations of rotors. Although accurate, the developed trajectory optimization method in [88] imposes prohibitive computational complexity for solving the trajectory optimization problem in real time, which is required for the autonomous navigation of the UAV. Additionally, the obstacle-free trajectory has not been considered, which makes the proposed technique ineffective for dense urban areas.

We have considered this requirement in our previous works and studied the kinematic-aware crash avoidance RISEUAV trajectory design in dense urban environments for intelligent vehicles [41] and IoVs [33,136]. However, a generic approach is required with a linear kinodynamic model and constraints for computationally efficient and real-time trajectory optimization for autonomous navigation applications. To bridge the gap, we have developed a kinodynamic-modeling-based trajectory optimization for UAV navigation in dense environments [117].

The basic obstacle-free trajectory optimization problem for a UAV-BS while satisfying kinodynamic motion constraints is modeled as $\mathcal{P}2$ for each sampling time k :

$\mathcal{P}2$:

$$\begin{aligned} \text{OF2. } \min_{\mathfrak{U}[k]} & \underbrace{\Xi_{1_1} \|\mathfrak{U}[k]\|_2^2}_{\text{OF2}_1} + \underbrace{\Xi_{1_2} \sum_{IV=1}^{IoVs} \gamma_{UAV-IV}^{LoS} \|p^{UAV}[k] - p_{IV}^{IoV}[k]\|_2^2}_{\text{OF2}_2} \\ \text{s. t. } & \\ \text{C1}_1. & \quad X[k+1] = GX[k] + H\mathfrak{U}[k] \\ \text{C1}_2. & \quad \xi[k] \cap \mathbb{P}_{\text{Obstacle}} = \emptyset \\ \text{C1}_3. & \quad Z_{\min} \leq z[k] \leq Z_{\max} \\ \text{C1}_4. & \quad v_{xy}[k] = \|v_x[k], v_y[k]\|_2 \leq V_{\max}, \quad u_z[k] \leq (U_{\max} - \gamma_1 v_{xy}[k]); \\ \text{C1}_5. & \quad \|a\|_2 \leq \mathcal{A}_{\max}; \\ \text{C1}_6. & \quad \psi[k] \leq \psi_{\max} \end{aligned}$$

where $\mathfrak{U}[k] = [u_1[k], u_2[k], u_3[k]]^\top$ denotes the input jerks to the UAV motion control system; the objective function OF2_1 minimizes the energy effort for navigation; objective function OF2_2 minimizes the average distance between UAV position at sample time k (i.e., $p^{UAV}[k]$) and positions of the intelligent vehicle IV in IoVs (i.e., $p_{IV}^{IoV}[k]$, $\forall IV \in \text{IoVs}$) that have valid LoS links with the UAV, where $\gamma_{UAV-IV}^{LoS} \in \{0, 1\}$ is the binary variable which is 1 for valid LoS links and 0 otherwise; C1_1 models kinodynamic equation of motion; C1_2 satisfies obstacle-free navigation, where $\mathbb{P}_{\text{Obstacle}}$ denotes the 3D coordinates of obstacles (i.e., buildings and skyscrapers); C1_3 limits the UAV altitude; $\text{C1}_{4,5}$ impose speed and acceleration constraints; and C1_6 imposes heading change nonholonomic constraint, where $\psi[k] = \tan^{-1}(v_y[k]/v_x[k])$, with $v_x[k]$ and $v_y[k]$ denoting the linear horizontal velocity along the X and Y axes, respectively. The following points can be observed from $\mathcal{P}2$.

1. $\mathcal{P}2$ is non-convex for C1_2 and C1_6 . The solutions are discussed in Section 6.
2. The motion primitives in $\mathfrak{U}[k]$ can be input speed or acceleration based on the density of the environment, see [117]. For example, as per Figure 4a, the input can be given as $\mathfrak{U} = [u, v_x = v \cos(\theta), v_y = v \sin(\theta)]^\top$.
3. A linear kinodynamic model should be used for modeling C1_1 to make the trajectory optimization problem computationally feasible for real-time, particularly for use in the context of model predictive control [117].
4. In the context of wireless communication support for IoVs in dense urban environments, modeling probabilistic LoS links modeled in (22) is not effective and reliable for safe autonomous driving. Therefore, deterministic LoS links modeling is considered in the optimization problem by embedding $\gamma_{UAV-IV}^{LoS} \in \{0, 1\}$, which makes the trajectory optimization a mixed-integer programming [93] and NP-hard optimization problem for modeling LoS links, which is highlighted in Lemma 1.
5. OF2_2 minimizes the UAV-IoVs distances which based on the channel model in (2), maximizes the channel gain. However, for the RISEUAV, based on the channel model in (9), the cascaded double LoS paths should be used as follows:

$$\text{OF2}_3 = \sum_{n=1}^N \sum_{IV=1}^{IoVs} \left(\left(\gamma_{RU-X_n}^{LoS} \|p_n^X[k] - p^{RU}[k]\|_2^2 \right) \left(\gamma_{RU-IV}^{LoS} \|p^{RU}[k] - p_{IV}^{IoV}[k]\|_2^2 \right) \right), \quad (24)$$

or

$$\text{OF2}_4 = \sum_{IV=i}^{IoVs} \sum_{IV=j}^{IoVs} \left(\left(\gamma_{RU-IV_i}^{LoS} \|p_i^{IoV} - p^{RU}[k]\|_2^2 \right) \left(\gamma_{RU-IV_j}^{LoS} \|p^{RU}[k] - p_j^{IoV}[k]\|_2^2 \right) \right), \quad (25)$$

where OF2_3 can be used for RISEUAV (RU), providing aerial CLoS link for V2X communication (with $p_n^X[k]$ denoting the position of X that can be BS, pedestrian, ...); OF2_4 can be used for RISEUAV (RU) to facilitate V2V communication [94].

6. In the case of the RISEUAV, where OF2₃ and OF2₄ may be used, the objective function is nonconvex, which is discussed in the solution in Section 6.
7. In formulating the *min* optimization problem $\mathcal{P}2$, we have used the following technique. Instead of using a *max* problem for achievable rate maximization in (9) and (10) with double CLoS path model in the denominator, we used the *min* optimization problem by using OF2₂, OF2₃, and OF2₄ with the double CLoS path model in the nominator. These two *min* and *max* problems are equivalent, see the solutions in Section 6.
8. The optimization variables in $\mathcal{P}2$ are input jerks (motion primitives) to the UAV motion system. Therefore, they can be directly used for UAV navigation without the requirement of an extra path-tasking or path-following controller.
9. Constraints related to channel performance and QoS (e.g., the minimum achievable rate for each UE or IoV) are not considered in $\mathcal{P}2$. However, it is typical in alternating optimization methods (such as SCP and SCA) for problem objectives to be solved iteratively by relaxing other objectives or constraints.

Lemma 1. *UAV trajectory optimization with valid LoS paths between UAV and target nodes, which are randomly obstructed (e.g., by building), is an NP-hard problem for convex programming.*

Proof of Lemma 1. When UAV waypoints experience LoS link obstructions due to obstacles (e.g., buildings), these obstructions can be modeled as fictitious obstacles that must be avoided. However, validating LoS links in such scenarios is computationally expensive. Specifically, it involves processes such as linear interpolation and checking whether the interpolated points lie within the set of obstacles, $\mathbb{P}_{\text{Obstacle}}$. Furthermore, the UAV's goal position is uncertain and depends on the movements of IoVs (ground vehicles). This means that the optimal UAV trajectory is dynamically affected by IoV movements, as described in OF2_{2,3,4}. As a result, the validation process requires checking an infinite number of potential (random) paths, which leads to the NP-hardness of the trajectory optimization problem in the context of wireless communication support for IoVs in future B5G and 6G networks. $\square$

Remark 11. *Lemma 1 highlights an open research question that has not been thoroughly investigated in the literature.*

5. Applications and Services

In this Section, we explore RIS-assisted, UAV-enabled communication from the perspectives of applications, services, aerial packages, UAV design, and RIS technologies. We briefly highlight some works and their methods, with a comprehensive list of relevant studies provided in Table 1.

5.1. Services

One of the promising applications is utilizing UAVs as aerial support for ground vehicles in dense environments, heavy roads, and traffic jams. In traffic monitoring, the UAV-BS can be deployed for malfunction or congestion relaxation for terrestrial cellular networks. However, the ground backhauling through direct LoS links for UAV-BSs may be randomly blocked by buildings due to the UAV-BS movement, whereas the backhaul rate must be high enough to support the traffic. Therefore, aerial backhaul system has been proposed through a high-altitude aerial platform (HAP) balloon RISEUAV [95], as shown in Figure 6. The passivity of the RIS element facilitates energy-efficient aerial LoS backhauling. The other advantage is that the aerial balloon RISEUAV can provide isotropic ($0 \sim 360^\circ$) reflection to cover a wider area, which is energy efficient from the UAV propulsion perspective. The 3D placement as well as phase alignment of aerial RISs are optimized so that the source transmit power is minimized. The optimization problem is solved through alternating optimization and using Weiszfeld's algorithm [137].

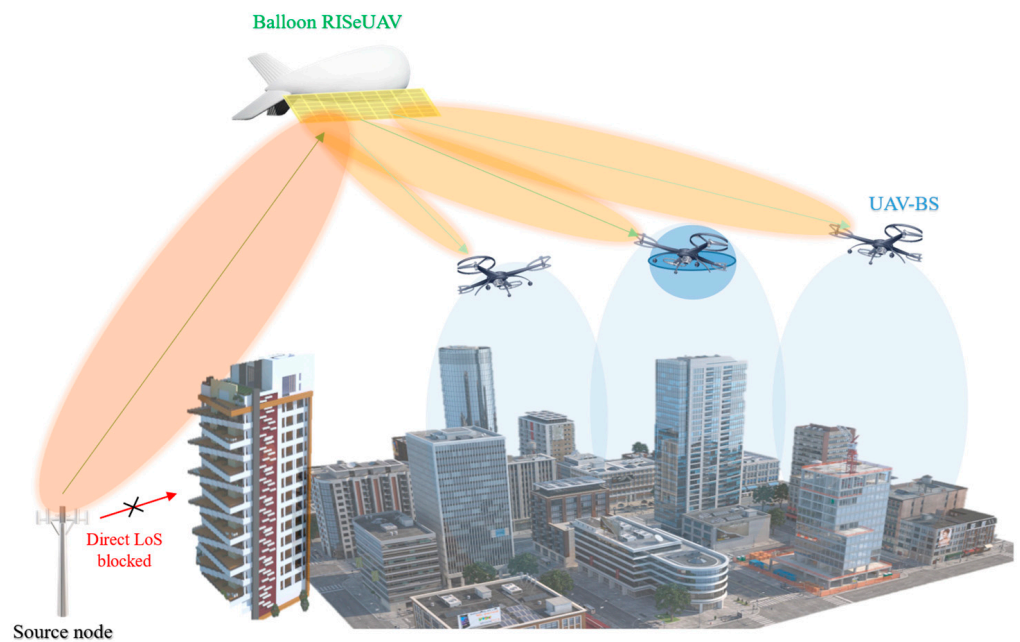


Figure 6. Aerial backhauling through the RISeUAV to UAV-BSs.

As another application, the UAV-relayed, RIS-assisted link was used to enhance satellite-terrestrial communication [138], accounting for hardware impairments and interference when direct LoS links were unavailable. A closed-form expression for the outage probability was derived to assess the impact of RIS and key parameters on system performance. RISeUAV also has been utilized for covert communication [109] to transmit information secretly.

5.1.1. IoTs Sensor Networks

UAVs are effective for enabling timely and reliable data delivery in remote IoT sensor networks. RISeUAV has been used to minimize the average age of information (AoI) of the IoT data received by the BS over time [103]. A time-constrained model-predictive-control-based algorithm, with a mathematically rigorous proof of the optimality, has been proposed for 3D UAV navigation for RIS-assisted data collection from sensor networks on uneven terrain [139].

5.1.2. Internet of Vehicles (IoVs)

Machine-type communications (MTCs) aim to establish a wireless network with low latency, high reliability, energy efficiency, and scalability to support the upcoming IoT applications [99]. The most promising application of UAV-enabled, RIS-assisted wireless communication is for intelligent vehicles and IoVs [136,140]. Enabling reliable V2X connectivity with low-latency communication is a big step toward realizing autonomous driving in smart cities [32,141,142].

The multi-UAV trajectory design has been studied for providing a quick wireless connection for IoVs [100]. The proposed techniques use SCA to handle nonconvexity and decompose the problem into three communication scheduling, power allocation and UAV trajectory optimization sub-problems. A conformal RIS array is mounted on ground vehicles as a relay [107]. To handle dynamic beamforming and complexities associated with adaptively controlling intelligent reflecting surfaces, DRL techniques have been used.

5.1.3. Mobile Edge Computing

UAV-enabled, RIS-assisted communication has been recently applied in the field of MEC [143]. A RISeUAV-assisted MEC is deployed to assist the communication between the

ground users [97]. The joint UAV trajectory, RIS passive beamforming and MEC resource allocation design are developed to maximize the energy efficiency of the system. To tackle the intractable non-convex problem, this is decomposed into two subproblems and solved iteratively based on SCA and the Dinkelbach method. The same study is conducted for the UAV-BS MEC assisted with ground RIS [51].

5.2. UAV Design

Depending on the applications, communication technologies, available resources, and UAV navigation objectives, the aerial UAV package can be designed as an aerial base station (UAV-BS) or equipped with an RIS to form a RISEUAV.

5.2.1. UAV-BS

Probably the common form of UAV-enabled wireless communication is utilizing UAVs as airborne BS. See Table 1 for the paper that studied the deployment of UAV-BS. However, the application of UAV-BS as active aerial BS can be computationally prohibitive, expensive, and inefficient in future B5G and 6G networks [144]. Because equipping UAVs with massive MIMO antennas for beamforming capability requires complex baseband processing and channel estimation. This seems impractical, as it makes the aerial package expensive and heavy. Notably, it is against UAV agility which is required to provide aerial LoS communication for intelligent vehicles [145]. Therefore, equipping UAVs with RIS as RISEUAV can be an alternative and cost-effective solution [41].

5.2.2. RISEUAV

RISeUAV, as an emerging integrated approach, has gained significant popularity in the community due to its ability to combine the advantages of both UAVs and RIS [145,146]. RISEUAV can be utilized in various applications such as joint wireless power transfer (WPT) and wireless information transfer (WIT) [83], for ergodic communication [147], mobile ground nodes [148], IoTs [149], vehicles [93], multi-layer MEC [150], disaster management [121], anti-jamming maritime [113], free-space optical communications [151], and aerial LoS links for intelligent vehicles [33,41,136]. See Table 1 for more papers that have used RISEUAV.

A theoretical model for analyzing the uplink performance of RISEUAV communication systems with a random 3D mobile pattern is presented in [152]. In this model, a random 3D mobility pattern for UAVs is considered, incorporating both random waypoint and uniform mobility models. Subsequently, an end-to-end transmission model for the RISEUAV communication system is developed, accounting for the effects of channel fading, RIS configuration, UAV mobility, and association policies. By combining these models, analytical expressions for uplink transmission metrics are derived, along with a performance-bound analysis.

The RISEUAV may experience random fluctuations due to wind disturbances and propeller rotations, leading to misalignment of reflected signal phases [54]. A 3D RISEUAV channel model, proposed in [153], simulates UAV fluctuations using random processes. Key statistics, including the temporal correlation function (TCF), Doppler power spectral density (PSD), and ergodic achievable rate, are derived. The results indicate that even minor vibrations can introduce additional Doppler frequency components, significantly reducing the temporal correlation and achievable rate of the RISEUAV-assisted channel, particularly when operating at high-frequency bands.

The RISEUAV may experience random fluctuations due to wind disturbances and propeller rotations. This leads to the misalignment of reflected signal phases [54]. A 3D RISEUAV channel model has been proposed in [153] where UAV fluctuations are simulated by random processes. Statistics including temporal correlation function (TCF), Doppler power spectral density (PSD), and ergodic achievable rate are obtained. Results show that even for a minor vibration, the UAV wobbling can introduce extra Doppler frequency

components and significantly reduce the temporal correlation and achievable rate of the RISEUAV-assisted channel when systems operating at high-frequency bands.

5.2.3. Solar-Powered UAVs

Toward more sustainable and carbon-neutral environments, solar energy has been extensively utilized in the energy sector [154], as well as in solar-charged batteries [155]. Solar-powered UAVs are expected to see increased usage in the future due to advancements in solar panels and microcontrollers [156]. The solar-powered RISEUAV has also garnered attention in the wireless communication community as a means of providing supplementary propulsion power [157]. The optimal joint active and passive beamforming, along with the energy-constrained RISEUAV trajectory, is derived in closed form, while also optimizing the UAV's size, weight, power limitations, reflecting elements, and flying/hovering time.

5.3. RIS Technologies

Different RIS technologies and variant types have been studied in the literature.

5.3.1. Intelligent Omni Surface

Different from the RIS which only reflects signals to the same side of the incident signal, the intelligent omni-surface (IOS) can also reflect and refract the signals to both sides of the surface [158], thereby providing full-dimensional rate enhancement [159,160]. The joint UAV trajectory and phase shift of IOS has been optimized in [96,105] by maximizing the achievable rate in the context of MEC.

5.3.2. STAR-RIS

Unlike conventional RIS, which only reflects incident signals, STAR-RIS transmits and reflects signals, enabling 360-degree coverage but requiring more complex coefficient design. In [98], a sum-rate maximization problem is addressed through joint optimization of UAV trajectory, active beamforming at the UAV, and passive beamforming at the STAR-RIS, using RL for decision-making to manage unknown obstacles. To bolster system robustness against environmental uncertainties, a distributionally robust RL (DRRL) algorithm is introduced, providing reliable worst-case performance. Results show that STAR-RIS enhances sum-rate performance compared to traditional RIS and that DRRL ensures more stable and robust outcomes than existing RL methods.

5.3.3. Holographic Surface

The reconfigurable holographic surface (RHS) is a type of RIS [161]. The RHS performance is very similar to that of RIS, with two major differences in the operation and application: (1) RHS consists of metamaterial elements with tunable radiation amplitude, which makes it feasible to tune the radiation pattern of the planar surface. Therefore, beamforming is implemented by controlling the radiation pattern of RHS elements rather than controlling the phase shifts [162], which makes it energy-efficient. (2) The RHS can function as a transceiver rather than only reflecting signals. UAV may be equipped with RHS as RHSeUAV that propagates communication signals generated by the UAV rather than reflecting external signals [12]. This results in two main advantages: (1) compared with UAV-BS, the onboard package of the RHSeUAV is energy efficient as the RHS adjusts the radiation patterns through the holographic surface that obviates active/passive beamforming [162]; (2) compared with the RISEUAV, the RHSeUAV's flight becomes independent of the ground MIMO BSs which adds more freedom for optimal navigation [163].

6. Solutions to Optimization Problems

In UAV-enabled, RIS-assisted communication, various objectives are considered, such as UAV placement and trajectory design, UAV energy consumption for communication and beamforming, BS power transmission, RIS ON/OFF status, phase shifts, and resource allocation. These objectives are typically formulated as optimization problems. Convex

programming is the most widely used solution due to its optimality and reduced time complexity. However, due to the complexity of the multilateral optimization problem, real-time solutions may not always be feasible, and the optimization process may converge to a sub-optimal solution after applying multiple convex relaxations. In such cases, heuristics and machine learning methods can be employed

6.1. Convex Programming

Convex programming is an effective technique for real-time optimization. However, nonconvexity in the optimization problems, including both objective functions and constraints, is common in UAV-enabled, RIS-assisted communication. As a result, alternating optimization (AO) and iterative optimization methods, such as SCP and SCA, are typically used to address the nonconvexity of these problems [100]. The nonconvexity in RIS-assisted UAV-enabled communication arises primarily from two sources: the channel modeling of the CLoS link and the nonholonomic constraints involved in trajectory optimization. Depending on the specific system and problem setup, additional nonconvexity terms may also be introduced into the optimization problem

To tackle the complexity, the optimization problem is normally decomposed into three sub-optimization problems: (P1) power allocation for communication; (P2) beamforming and phase shift design; and (P3) path-planning and trajectory optimization; in each of which other optimization variables are constant.

For example, by considering a fixed trajectory and phase shift design the power transmission problem P1 in the achievable rate formula in (10) is solved using a standard convex programming solver such as CVX [48]. Then, the phase shift design rule is given in Equation (8). However, the trajectory optimization problem P3 is nonconvex as we highlighted in Remark 3. To tackle this issue, slack convex objective function is utilized as per the convex relaxations technique presented in Lemma 2 given from [164].

Lemma 2. With given Q_1, Q_2, Q_3, τ and κ all positive values, function $f(x, y)$ is convex with respect to $x > 0$ and $y > 0$

$$f(x, y) = \log_2 \left(Q_1 + \frac{Q_2}{x^{\frac{\tau}{2}} y^{\frac{\kappa}{2}}} + \frac{Q_3}{x^{\tau} y^{\kappa}} \right), \quad (26)$$

See the proof in [164].

Since $f(x, y)$ is convex with respect to x and y , it can be linearized by applying the first-order Taylor expansion at a given point [164].

Other useful tools are the following inequalities:

$$\mathbf{a} \cdot \mathbf{b} \leq \|\mathbf{a}\| \|\mathbf{b}\| \quad (27)$$

$$\left(\mathbf{a}^2 + \mathbf{b}^2 \right)^{1/2} \leq (\mathbf{a} + \mathbf{b}) \quad (28)$$

For instance, to handle the nonconvexity of the optimization problem, after applying the phase shift rule (8), the equivalent formula of the achievable rate in (10) is obtained by using inequality (28) and applying Euler's formula in [96]. Then, by the first-order Taylor approximation, the convex model is developed and solved by a standard convex (CVX) solver. Also, the nonconvexity of the UAV linear speed optimization formula in (24) has been tackled by applying the inequality (29) in [97] to obtain the convex upper bound of (24).

Alternative methods are defining convex slack variables. For example, (9) is formulated as

$$\max g_{CLoS}^{TR} \approx \max(\alpha\beta), \quad (29)$$

where α and β are slack variables corresponding to $d_{T_RIS}^T$ and $d_{RIS_R}^{RIS}$ in (9), respectively. Then, the following constraints are applied:

$$\alpha \leq \underbrace{\|p^{Tx} - p^{RIS}\|_2}_{d_{T_RIS}^T} \quad \& \quad \beta \leq \underbrace{\|p^{RIS} - p^{Rx}\|_2}_{d_{RIS_R}^{RIS}}, \quad (30)$$

Then, it is proved that the relaxed recasting problem is equivalent to the original problem [94]. The maximization problem in (30) is globally lower-bounded by calculating the first-order Taylor expansion of (31) at the local point. The nonconvexity of the (30) itself can be handled by defining a new slack variable ζ and imposing additional constraint $\zeta \leq \alpha\beta$, which can be linearized by applying the logarithm function to the formula ($\ln \zeta \leq \ln \alpha + \ln \beta$) and linearizing it by using the first-order Taylor expansion [94].

The solution to nonconvexity of nonholonomic heading constraint C1₆ in $\mathcal{P}2$ can be found in Lemma 1 in [117].

6.2. Heuristics Algorithms

Metaheuristic optimization methods, such as evolutionary algorithms including genetic algorithms (GA), particle swarm optimization (PSO), quantum PSO (QPSO), and simulated annealing (SA), are popular techniques in the engineering field [165–167]. These methods are useful for solving optimization problems that are highly nonlinear, nonconvex, and complex. They can also be applied when the system's mathematical model is unclear or too complex to develop.

PSO and SA have been used in a hybrid algorithm for RISEUAV deployment and RIS elements allocation [70]. The authors of [75,168] have used the PSO technique as the benchmark to evaluate their proposed RISEUAV path planning method. Further, GA has been used to solve subproblems of RISEUAV path planning [80]. To tackle the computational complexity of the multilateral optimization problem the blocked coordinate descent (BCD) alternating optimization approach is used. The optimization problem is regarded as mixed integer nonlinear programming (MINP), and different objectives are solved using a combination of heuristic and gradient-based techniques. A combination of binary and continuous GA is used to solve the UAV beamforming and RIS ON/OFF and phase shift optimization problem.

Although heuristic optimization methods are easy to formulate and solve, they suffer from the risk of being trapped in sub-optimal solutions and require extremely long runtimes. Additionally, these methods lack learning capabilities, making them unsuitable for real-time optimization. As a result, machine learning (especially deep reinforcement learning, or DRL) techniques have gained popularity in recent years.

6.3. Machine (Deep) Learning

Machine learning and AI-assisted techniques have gained increased attention for handling the nonlinearity, nonconvexity, and computational complexity of complex engineering problems [169]. In addition to the nonconvex formulation associated with UAV-enabled, RIS-assisted communication and trajectory optimization, challenges such as imperfect CSI, dynamic environments, and uncertain channel conditions further complicate the problem. As a result, machine learning techniques have been increasingly considered in the wireless communication community.

Aiming to improve the performance of RISEUAV communication channels, the authors in [170] proposed a machine learning-based beamforming policy for UAV-RIS by employing prioritized experience replay (PER) based deep Q-Network (DQN). Further, they used a DDPG-based beamforming approach for secure UAV-RIS beamforming under imperfect channel estimation [171].

Deep Reinforcement Learning (DRL)

Among AI-based machine learning techniques, DRL is the most popular one in the community, see Table 1. DRL works based on Markov decision process (MDP) and dynamic programming for estimating future reward in the MDP for training DQN [136].

The most common DRL agent is the actor-critic deep deterministic policy gradient (DDPG) agent. A schematic of the DDPG DRL agent including actor-critic deep networks is illustrated in Figure 7. Based on the MDP, the actor network π_{ϕ_A} receives the current state $\mathcal{S}[k]$ (e.g., positions of the RISEUAV and IoVs) and takes action $\mathcal{A}[k] = \pi_{\phi_A}(\mathcal{S}[k])$ in the permitted action space (e.g., the action may include UAV motion speed inputs or motion primitives as well as RIS-PhSh). Based on the taken action $\mathcal{A}[k]$, the rewards $\mathcal{R}[k]$ and new state $\mathcal{S}[k+1]$ are determined based on the simulated environment. For example, the RISEUAV takes the given action and moves through the path or generated trajectory. Then, the channel gain and communication performance indexes are obtained to calculate the reward. The new state $\mathcal{S}[k+1]$ is the new positions of the RISEUAV and IoVs. The critic networks $Q_{\phi_c}(\mathcal{S}[k], \mathcal{A}[k])$, is embedded to predict future rewards, and is trained by minimizing the error function of the calculated actual reward $\mathcal{R}[k]$ and its estimated rewards $Q_{\phi_c}(\mathcal{S}[k+1], \pi_{\phi_A}(\mathcal{S}[k+1]))$. Therefore, critic $Q_{\phi_c}(\mathcal{S}, \mathcal{A})$ is a Q-value function based on the recursive Bellman equation:

$$Q_{\phi_c}(\mathcal{S}[k], \mathcal{A}[k]) = \mathbb{E}[\mathcal{R}[k] + \gamma Q_{\phi_c}(\mathcal{S}[k+1], \pi_{\phi_A}(\mathcal{S}[k+1]))] \quad (31)$$

where γ is the discount factor for the future award, and $\mathbb{E}$ denotes expectation.

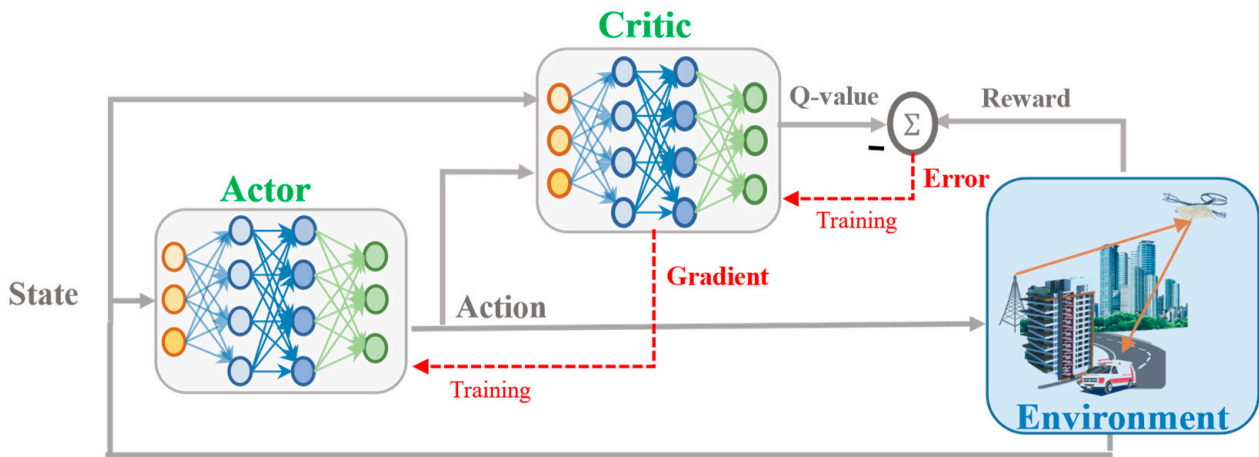


Figure 7. The schematic of the actor-critic deep deterministic policy gradient DRL agent.

The critic parameters ϕ_c are updated by minimizing the loss function L given by all sampled experiences out of M random mini-batch of experiences (trial-and-error episodes)

$$L = \frac{1}{M} \sum_{i=1}^M (\mathcal{Y}_i[k] - Q_{\phi_c}(\mathcal{S}[k], \mathcal{A}_i[k]))^2 \quad (32)$$

where

$$\begin{aligned} \mathcal{Y}[k] &= \mathcal{R}[k] + \gamma \mathbb{E}(\mathcal{R}[k+1]) \\ &= \mathcal{R}[k] + \gamma Q_{\phi_{ct}}(\mathcal{S}[k+1], \pi_{\phi_{At}}(\mathcal{S}[k+1])); \end{aligned} \quad (33)$$

The actor parameters ϕ_A are updated using the sampled policy gradient by applying the chain rule to the expected return that maximizes the expected discounted reward:

$$\begin{aligned} \nabla_{\phi_A} J &= \frac{1}{M} \sum_{i=1}^M \nabla_{\phi_A} Q_{\phi_c}(\mathcal{S}[k], \pi_{\phi_A}(\mathcal{S}[k])); \\ &= \frac{1}{M} \sum_{i=1}^M \nabla_{\pi} Q_{\phi_c}(\mathcal{S}[k], \pi_{\phi_A}(\mathcal{S}[k])) \nabla_{\phi_A} \pi_{\phi_A}(\mathcal{S}[k]) \end{aligned} \quad (34)$$

However, the design of the DRL structure, including the deep networks and the architecture of the actor-critic, the method of observing the system state, the action-taking process, reward functions, and the training algorithm of the deep neural network, all significantly impact the performance of the DRL agents. In [136] we have trained two DRL agents, i.e., RU-LoSNet and RU-PixelNet for the trajectory design of RISEUAV for IoVs in dense urban areas. The RU-PixelNet, with convolutional neural networks (CNN) architecture, showed promising performance by receiving pixels of aerial images as the input. Pixels indicate the IoVs' positions and RISEUAV's current position as the state input. However, it was trained offline by developing the occupancy map of the dense urban area and detailed channel modeling.

Nevertheless, there are critical issues with the DRL agent. Its dependency on MDP and randomness of the environment (e.g., random obstruction of LoS paths) makes the training process challenging. Specifically, frequent terminating the trial-and-error episodes, due to violating constraints, may make the training process unstable. Tuning the algorithm and hyperparameters to achieve stable results may lead to a conservative non-optimal solution. Soft actor-critic (SAC) DRL agent seems to be the most effective agent by considering the policy entropy (i.e., policy uncertainty) to promote more exploration while training the actor-critic in a stochastic environment [136]. Another problem is that the DRL training is based on dynamic programming to estimate future commutative rewards, see (32). On the other hand, the UAV's final position is uncertain depending on the IoVs routes. Therefore, training DRL agents as UAV path planners in the Markovian process of a random environment can be problematic.

As an alternative solution, for the first time, we have proposed training generative adversarial networks (GANs) offline to generate random trajectories in real-time [172]. GANs are very popular in computer and data science to create fake data (e.g., images) when the source data are not sufficient for training. We have used this feature for real-time trajectory generation for RISEUAV or UAV-BS in dense urban environments. Since GANs do not rely on MDP, the GANs path planner can achieve more optimal results compared to DRL agents.

Federating learning is a popular technique for training multi-DRL agents where data privacy is a matter of concern [173].

6.4. Game Theory

The joint communication performance and energy-efficient (RISeUAV or UAV-BS) trajectory optimization can be regarded as the (bandit or Stackelberg) game theory approaches [91,174]. Because the UAV may navigate more spaces to improve communication performance, it consumes more energy for propulsion. Other game players can be bandwidth split, energy trading for communication, RIS allocation, etc.

The RISEUAV navigation for WiGig communication is modeled as a contextual multi-armed bandit (MAB) game [91], where the UAV acts as the learner. It aims to maximize its rate (reward) by serving various trajectory hotspots, considered as the bandit's arms. The joint optimization of received signal strength at the UAV and users' energy efficiency is modeled as a single-leader/multiple-followers Stackelberg game [175]. The UAV, as the leader, optimizes phase shifts for improved signal quality, while users, as followers, engage in a non-cooperative game to maximize their energy efficiency by adjusting uplink transmission power.

In RIS-assisted UAV communication for IoT sensor networks, IoT devices use energy from various providers, prompting an energy trading mechanism via a hierarchical Stackelberg game [92]. IoT devices aim to maximize their rate benefit minus payment for wireless charging, while energy providers balance revenue and energy costs. To assess fairness, objective functions based on sum-rate and minimum-rate maximization are defined, optimizing UAV trajectory, BS transmit power, and RIS phase shift.

6.5. Time (Computational) Complexity

An important consideration in the trajectory optimization of UAVs for IoVs is the time complexity associated with the formulated optimization problem and its solution. UAVs are designed to navigate autonomously and support IoT and IoV systems [117]. This necessitates that the optimization process be executed in real-time to enable UAVs to perform their missions autonomously, without reliance on external computation or memory resources. While convex programming solutions offer the advantage of real-time implementation, they do so at the expense of solution optimality. As demonstrated by Lemma 1, trajectory optimization involving valid LoS paths is NP-hard, making real-time solutions computationally prohibitive.

Sampling-based techniques, such as rapidly exploring random trees (RRT), are widely used in the robotics field for trajectory optimization in complex environments [148]. However, the computational cost of these methods is significant when applied to wireless communication scenarios, where each sample must be evaluated against multiple objectives. This further complicates their real-time applicability, especially for dense urban environments where maintaining LoS links for IoV support is crucial. Therefore, AI-based agents, if properly trained offline, can be the best candidates for real-time trajectory generation and communication control. Alternatively, they can be deployed to address the nonconvexity of the optimization problem, thereby overcoming the limitations of convex programming, such as sub-optimality.

7. Integrated Localization and Communication

This section focuses on the importance of localization for communication and autonomous navigation of unmanned vehicles and studies the joint localization and communication in future mmWave networks [176,177].

7.1. Joint Localization and Communication

From the wireless communication perspective, localization is important for improved connectivity and communication [177]. Localization of UEs helps in CSI estimation [178], UE allocation to cellular networks and avoiding intercell interference, secure communication against eavesdropping, etc. [179]. Notably, lately, position-aware beamforming with directional beam-steering is essential for dynamic environments with mobile UEs such as intelligent vehicles and vehicular networks [180]. Further, object sensing and detection using RF signals [181] is a practical technology as being used in radar sensors in intelligent vehicles [182]. In this context, the integrated sensing and communication (ISAC) concept has been introduced to maximize the energy and spectral efficiency of wireless channels [183].

On the other hand, from the vehicular navigation perspective, accurate localization is critical for autonomous navigation and unmanned driving for intelligent vehicles [184,185]. In outdoor environments, localization is implemented with the aid of a global positioning system (GPS). However, the meter-scale accuracy of the GPS signals is not sufficient for the safe driving of unmanned vehicles. Particularly, GPS signals are not available in indoor areas and their interruptions in dense environments, such as urban areas, pose fatal autonomous-vehicles safety risks. Therefore, sensor fusion based localization is performed to achieve more robust and reliable localization for safe unmanned driving [185].

Autonomous mobile robots and vehicles normally rely on onboard sensors for localization such as inertial navigation systems and odometry, radar, light detection and ranging (LiDAR), and cameras, among others. To minimize localization errors, the simultaneous localization and mapping (SLAM) concept has been very popular in the robotic community [186]. In SLAM, by combining the localization information given by the inertial navigation system (odometry) with the sensing and measurement data given by sensors, the localization error due to odometry drift or sensor inaccuracy is minimized. Also, simultaneously, a map of the environment is created. Lidar-based SLAM, although more accurate, is expensive and suffers from its dependency on mechanical processes to send

light signals in the 360-degree direction. Recently, visual SLAM using pixels has become very popular, particularly in autonomous vehicles, thanks to the developments in computer vision and deep learning techniques for extracting geometry information from pixels [187]. Nevertheless, still, visual-based localization is not robust [188], as it is impacted by weather conditions and technical aspects such as depth detection [189].

7.2. RIS-Assisted mmWave Localization

Therefore, mmWave signals, which are ubiquitous in future wireless networks, can be used for accurate localization as complementary to visual SLAM in the robotic community for autonomous vehicles [190]. In this light, MIMO beamforming for sending narrow-beam directional waves with angular resolution is very effective in promoting mmWave-based localization [191]. In this context, the RIS technology is effective in enhancing energy-efficient mmWave localization [192]. For example, the combination of MIMO-active beamforming and RIS-passive beamforming for vehicular localization is studied [193]. Following the employment of RIS for localization, the meta-radar and meta-SLAM concepts have been proposed [194]. Furthermore, RIS-assisted mmWave radar SLAM has been proposed for autonomous vehicles [195].

Many works can be found in the wireless community literature that studied RIS-assisted localization for stationary and quasi-stationary UEs [196,197]. However, the application of RIS for localization in dynamic environments with UAVs and IoVs is a trending research topic [198,199], particularly by using deep learning techniques to tackle practical problems [200].

7.3. Localization for RISeUAVs

The localization-oriented study for UAV-enabled, RIS-assisted communication can be considered from two perspectives:

- (1) Positioning-aware beamforming of the aerial RIS in the RISeUAV.
- (2) Accurate localization of RISeUAV and ground vehicles.

7.3.1. Positioning-Based Beamforming

As highlighted in Remark 5, beamforming optimization and CSI estimation in dynamic environments are computationally prohibitive. Notably, it is difficult to retain beam alignment for moving vehicles as the channel state is changing not only due to the vehicle movement but also due to changing the signature of the environment. Therefore, the CLoS links in the BS-RISeUAV-UE paths (e.g., in an indirect V2V link through RISeUAV) are fragile and prone to instability. Therefore, positioning-based beamforming (i.e., directional beam-steering) through codebook design has been suggested [63,201].

7.3.2. RISeUAV Localization

RISeUAV is energy-efficient and cost-effective for various applications as we have discussed in this paper. Also, the RISeUAV can help ground vehicles with accurate localization and communication [202]. However, for RISeUAV to accomplish its tasks, particularly for aerial RIS phase shift design, the UAV's accurate position must be known, which is a non-trivial task. As stated, GPS signals are not accurate for RISeUAVs for beam-steering for autonomous crash-avoidance flights in dense environments. To this end, we have proposed the SLAPS technique [46] for simultaneous localization and phase shift of RISeUAV. In the SLAPS, the localization estimation and RIS phase shift are iteratively modified and optimized so both converge to the optimal solution, which is accurate localization-based communication. The SLAPS mechanism has been illustrated in Figure 8.

The SLAPS principle is as follows: Suppose that the actual RISeUAV pose (i.e., position and heading) is denoted with $\hat{x}$. The mMIMO BS beamforming and RIS-PhSh are implemented based on the RISeUAV's estimated pose (denoted as x). Then, the estimated pose and consequence BS-BF and RIS-PhSh are iteratively updated and optimized so that

the received signal at the receiver (e.g., an anchor point) is maximized, indicating that x converges to $\hat{x}$, where the maximum channel gain is achieved.

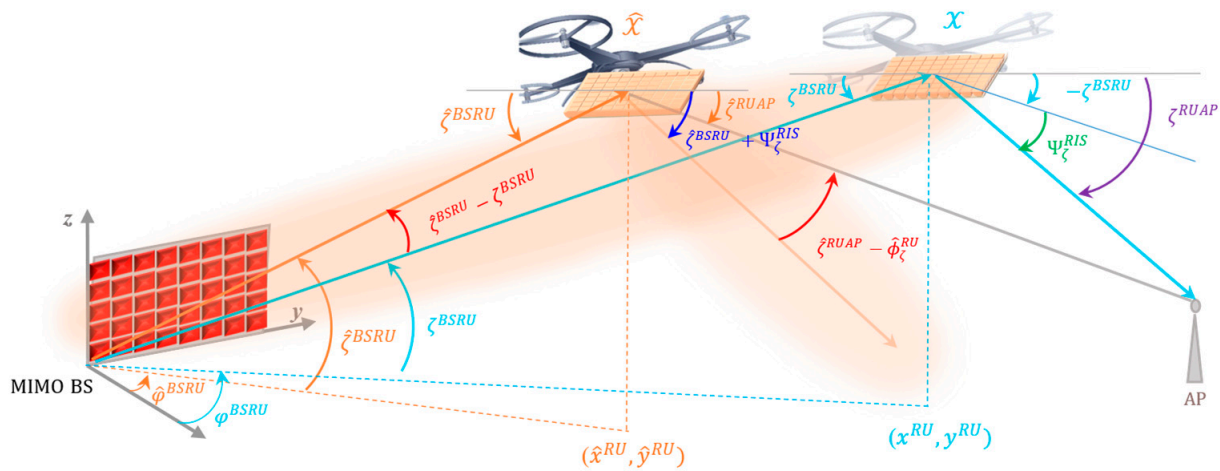


Figure 8. The geometry of the SLAPS for RISeUAV.

8. Future Research Directions

Despite significant advancements in UAV and RIS-assisted wireless communication systems, several areas require further exploration to fully realize the potential of these technologies in future B5G/6G networks. Below, we outline key open research topics:

1. **Obstacle-free trajectory optimization in smart cities:** Navigating complex, obstacle-rich environments in smart cities demands advanced optimization techniques. Future research should focus on developing robust algorithms that ensure obstacle-free trajectories while maintaining energy efficiency and communication reliability. Addressing this challenge is vital for safe and seamless UAV operations in urban settings.
2. **Channel modeling for RISeUAV systems:** The impact of UAV fluctuations due to disturbances such as wind or misalignment must be accounted for in channel modeling for RIS-assisted UAVs. Developing accurate models that consider these variables and incorporate UAV motion primitives adaptable to UAV dynamics will improve the resilience and effectiveness of communication links.
3. **AI-based path planning with visual inputs:** While AI and machine learning techniques have been employed for UAV path planning, more attention is needed on learning-based end-to-end methods [203], particularly those leveraging visual inputs. These approaches can enable UAVs to make complex navigation decisions in real time, which is crucial for applications in dense urban environments where dynamic obstacles and complex terrains are common.
4. **Modern AI and transformer neural networks for trajectory optimization and RIS control:** The use of modern AI architectures, such as transformer neural networks and visual transformers (ViT), present promising avenues for optimizing UAV trajectories and RIS phase shift control. Transformers can process large datasets and capture long-term dependencies effectively, making them suitable for real-time decision-making in UAV navigation and RIS configuration.
5. **Integrated localization and communication:** The integration of localization and communication functions is critical for enhancing the reliability and efficiency of UAV-based systems. Future works should explore methods to synergize these functions, ensuring accurate positioning while maintaining robust communication links. Open research areas include joint algorithm development, hardware design, and real-time implementation strategies.
6. **Computationally feasible solutions for autonomous navigation:** Developing algorithms that balance computational efficiency with performance accuracy is essential

for enabling real-time autonomous UAV navigation. Current methods often struggle to incorporate kinodynamic and motion constraints while maintaining obstacle-free trajectories, especially in complex urban environments. Future research should focus on creating lightweight, scalable algorithms capable of handling high-dimensional data and constraints without excessive computational demands. Strategies such as model predictive control with simplified dynamics [204], hybrid offline-online approaches, and leveraging advanced AI techniques could provide a path forward to achieving real-time, reliable autonomous navigation.

7. **Satellite-aided communication:** Leveraging satellite communications for enhancing connectivity in RIS-UAV-assisted IoT and vehicular networks should be explored. This includes multi-layered communication frameworks that combine terrestrial, aerial, and satellite links for improved coverage and reliability.
8. **Security and privacy concerns:** Ensuring the security and privacy of data transmission in UAV-assisted, RIS-enabled IoT and vehicular networks is another area that needs further investigation. Techniques for protecting against eavesdropping, jamming, and spoofing, as well as secure protocols for data exchange, remain essential topics for future research.

9. Conclusions

UAV-enabled, RIS-assisted wireless communication offers a transformative solution for enhancing the efficiency and reliability of future 5G and 6G networks, particularly for the Internet of Vehicles (IoVs). By harnessing the unique capabilities of UAVs and RIS, these systems effectively address mmWave signal propagation challenges, such as attenuation and blockage, while significantly improving spectral and energy efficiency. Insightful channel modeling, real-time energy-efficient UAV trajectory optimization, and innovative optimization techniques contribute to robust and sustainable communication networks. The diverse applications of UAV base stations and RIS-enabled UAVs in IoT sensor data collection, V2X communication, mobile edge computing, and smart city infrastructure demonstrate their versatility and essential role in modern wireless networks. Additionally, integrated localization and communication are vital for future dynamic vehicular networks, with UAVs and RISs overcoming practical barriers. Through this comprehensive review, this paper provided valuable insights into current directions, existing challenges, and emerging research topics, highlighting the potential of UAV-enabled, RIS-assisted communication systems to pave the way for more resilient and high-performance wireless networks.

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Abbreviations

AEH	Adaptive energy harvesting	MRT	Maximum ratio transmission
A2G	Air-to-ground	MTC	Machine-type communication
AoI	Age of information	NLoS	Non-line-of-sight
AO	Alternating optimization	NOMA	Non-orthogonal multiple access
AR	Arrey response	OF	Objective function
BCD	Blocked coordinate descent	PPO	Proximal policy optimization
BF	Beamforming	PSO	Particle swarm optimization
BS	Base station	P-BCD	Penalty-based block coordinate descent
CDF	Cumulative distribution function	RHS	Reconfigurable holographic surface
CLoS	Cascaded line-of-sight	RIS	Reconfigurable intelligent surface
DDPG	Deep deterministic policy gradient	RISeUAV	RIS-equipped UAV
DDQN	Double deep Q-network	RIS-PhSh	RIS Phase Shift
DFT	Discrete Fourier transform	RSMA	Rate-splitting multiple access
DQN	Deep Q-network	SA	Simulated annealing
DR	Data rate	SDP	Semi-definite programming
DRL	Deep reinforcement learning	SEE	Secrecy energy efficiency
EE	Energy-Efficiency	SIC	Successive interference cancelation
EH	Energy harvesting	SINR	Signal-to-interference-plus noise ratio
FFM	Fourier feature mapping	SCP	Sequential convex programming
FSPL	Free Space pathloss	SCA	Successive convex approximation
FHRE	Fronthaul-rate-ensuring	SDMA	Space division multiple access
HAP	High-altitude aerial platform	SNR	Signal-to-noise ratio
GA	Genetic algorithm	STAR-RISs	Simultaneously transmitting and reflecting RISs
GRP	Gaussian randomization procedure	SWIPT	Simultaneous wireless information and power transfer
G2G	Ground-to-ground	T-UAV	Tethered-UAV
GU	Ground users	UAV	Unmanned aerial vehicle
LAP	Ligh-altitude aerial platform	UAV-BS	UAV as BS
LoS	Line-of-sight	UE	User equipment
IOS	Intelligent omni-surface	ULA	Uniform linear array
MAB	Multi-armed bandit	UPA	Uniform planar array
MAT	Multiple antenna	WCSR	Worst-case secrecy rate
MDP	Markov decision process	WIT	Wireless information transfer
MEC	Mobile edge computing	WOA	Wales optimization algorithm
MIMO	Multi-input multi-output	WPT	Wireless power transfer
MINP	Mixed integer nonlinear program	WSDR	Weighted sum data rate
		WSR	Weighted sum-rate

Appendix A

The coordinates of m^{th} RIS element is calculated as follows (see Figure 4):

$$p_m^{RIS} = p^{RIS} + \Gamma^{-1}(\theta^{UAV}) \begin{bmatrix} l(m) \\ w(m) \\ 0 \end{bmatrix}, \quad (A1)$$

where

$$l(m) = d_l \left(\text{Mod}(m-1, M_l) - \left(\frac{M_l-1}{2} \right) \right);$$

$$w(m) = d_w \left(\left(\frac{M_w+1}{2} \right) - \left\lceil \frac{m}{M_l} \right\rceil \right);$$

$$\Gamma(\theta^{RU}) = \begin{bmatrix} \cos(\theta^{UAV}) & -\sin(\theta^{UAV}) & 0 \\ \sin(\theta^{UAV}) & \cos(\theta^{UAV}) & 0 \\ 0 & 0 & 1 \end{bmatrix};$$

M_l and M_w are the number of elements along with the length and width of the RIS, respectively; and d_l and d_w denote the length and width sizes of the elements, respectively. Also, $\Gamma(\theta^{UAV}) \in \mathbb{R}^{3 \times 3}$ transforms the coordinates from the global reference frame to the local reference frame of the UAV, and θ^{UAV} denotes the UAV heading with respect to the x-axis.

Appendix B

Assuming that 2D coordinates of the T_x is taken as the origin of the local reference frame, i.e., $p^{Tx} = [0 \ 0 \ z^{Tx}]^T \in \mathbb{R}^3$, the azimuth angle of the T_x – RIS path, is measured clockwise from the x-axis to the line between RIS position (i.e., p^{RIS}) and the T_x position, and is obtained as follows (see Figure 4):

$$\varphi^{T-RIS} = \cos^{-1}(x^{RIS} / \|[x^{RIS}, (y^{RIS})]^T\|_2); \quad (A2)$$

The elevation angle between the horizontal plane and T_x – RIS LoS link is obtained as

$$\zeta^{T-RIS} = \tan^{-1}((z^{Tx} - z^{RIS}) / \|[x^{RIS}, (y^{RIS})]^T\|_2); \quad (A3)$$

Similarly, the azimuth and elevation angles of the reflected signal from RIS can be obtained as

$$\varphi^{RIS-R} = \cos^{-1}\left(\frac{(x^{Rx} - x^{RIS})}{\|[x^{BS_{Rx}} - x^{RV}], (y^{BS_{Rx}} - y^{RV})]^T\|_2}\right); \quad (A4)$$

$$\zeta^{RIS-R} = \tan^{-1}\left(\frac{(z^{Tx} - z^{RIS})}{\|[x^{Rx} - x^{RIS}], (y^{Rx} - y^{RIS})]^T\|_2}\right). \quad (A5)$$

Appendix C

Supposing that the UPA of the mMIMO BS is aligned in the YZ plane (see Figure 5), the coordinates of the n^{th} antenna of the mMIMO BS is obtained as follows:

$$p_n^{BS} = p^{BS} + \begin{bmatrix} 0 \\ d_c \left(\text{Mod}(n, N_c) - \frac{1}{2} \right) \\ d_r \left(\left\lceil \frac{n}{N_c} \right\rceil - \frac{1}{2} \right) \end{bmatrix}, \quad (A6)$$

where N_r and N_c denote the number of horizontal and vertical antennas of the mMIMO BS; d_r and d_c denotes the horizontal and vertical sizes of each antenna (including the antenna spacing).

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